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Inventor(s)

Ganiger; Ravindra Shankar et al.

COUPLING ASSEMBLY FOR A TURBINE ENGINE

Abstract

A coupling assembly for a turbine engine. The coupling assembly includes a cold side component, a hot side component, and a fastening mechanism. The cold side component and the hot side component together at least partially form a combustion chamber. The fastening mechanism couples the hot side component to the cold side component. The fastening mechanism includes a stud disposed through the cold side component and a cap positioned on the stud. The cap defines a hollow interior and includes one or more first cap cooling holes. The one or more first cap cooling holes operably direct cooling air into the hollow interior such that the hollow interior provides a cushion of air between the combustion chamber and the stud.

Inventors: Ganiger; Ravindra Shankar (Bengaluru, IN), Nath; Hiranya (Bengaluru, IN), Benjamin; Michael A. (Cincinnati, OH), Brown; Daniel D. (Cincinnati, OH), Pal; Sibtosch (Mason, OH), Zelina; Joseph (Waynesville, OH)

Applicant: General Electric Company (Evendale, OH)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of U.S. patent application Ser. No. 18/314,565 filed on May 9, 2023, which claims the benefit of Indian Patent Application No. 202211060009, filed on Oct. 20, 2022, the contents of both of which are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

[0002] The present disclosure relates generally to coupling assemblies for turbine engines.

BACKGROUND

[0003] A turbine engine may include a combustion section having a combustor that generates hot combustion gases discharged into a turbine section of the turbine engine. The combustion section may include a cold side component and a hot side component that are coupled together by a coupling assembly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The foregoing and other features and advantages will be apparent from the following, more particular, description of various exemplary embodiments, as illustrated in the accompanying drawings, wherein like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.

[0005] FIG. 1 is a schematic cross-sectional diagram of a turbine engine, taken along a centerline axis of the turbine engine, according to the present disclosure.

[0006] FIG. 2 is a schematic partial cross-sectional view of a portion of an exemplary combustion section of the turbine engine of FIG. 1, according to the present disclosure.

[0007] FIG. 3 is a schematic partial cross-sectional view of a portion of a coupling assembly for the combustion section of the turbine engine of FIG. 1, taken at a centerline axis of the turbine engine, according to the present disclosure.

[0008] FIG. 4 is a top view of a washer isolated from the coupling assembly of FIG. 3, according to the present disclosure.

[0009] FIG. 5 is a schematic partial cross-sectional view of a portion of a coupling assembly for the combustion section of the turbine engine of FIG. 1, taken at a centerline axis of the turbine engine, according to another embodiment.

[0010] FIG. 6 is a schematic partial cross-sectional view of a portion of a coupling assembly for the combustion section of the turbine engine of FIG. 1, taken at a centerline axis of the turbine engine, according to another embodiment.

[0011] FIG. 7 is a schematic partial cross-sectional view of a portion of a coupling assembly for the combustion section of the turbine engine of FIG. 1, taken at a centerline axis of the turbine engine, according to another embodiment.

[0012] FIG. 8A is a schematic partial cross-sectional view of a portion of a coupling assembly for the combustion section of the turbine engine of FIG. 1, taken at a centerline axis of the turbine engine, according to another embodiment.

[0013] FIG. 8B is a top view of a plurality of flexible plates isolated from the coupling assembly of FIG. 8A, according to the present disclosure.

[0014] FIG. 9A is a schematic partial cross-sectional view of a portion of a coupling assembly for the combustion section of the turbine engine of FIG. 1, taken at a centerline axis of the turbine engine, according to another embodiment.

[0015] FIG. 9B is a top view of a plurality of flexible plates isolated from the coupling assembly of FIG. 9A, according to another embodiment.

DETAILED DESCRIPTION

[0016] Additional features, advantages, and embodiments of the present disclosure are set forth or apparent from a consideration of the following detailed description, drawings, and claims. Moreover, both the foregoing summary of the present disclosure and the following detailed description are exemplary and intended to provide further explanation without limiting the scope of the disclosure as claimed.

[0017] Various embodiments of the present disclosure are discussed in detail below. While specific embodiments are discussed, this is done for illustration purposes only. A person skilled in the relevant art will recognize that other components and configurations may be used without departing from the spirit and the scope of the present disclosure.

[0018] As used herein, the terms “first,” “second,” “third,” and “fourth” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.

[0019] The terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows. The terms “forward” and “aft” refer to relative positions within a turbine engine

[0020] or vehicle, and refer to the normal operational attitude of the turbine engine or vehicle. For example, with regard to a turbine engine, forward refers to a position closer to an engine inlet and aft refers to a position closer to an engine nozzle or exhaust.

[0021] The terms “coupled,” “fixed,” “attached,” “connected,” and the like, refer to both direct coupling, fixing, attaching, or connecting, as well as indirect coupling, fixing, attaching, or connecting through one or more intermediate components or features, unless otherwise specified herein.

[0022] The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

[0023] As used herein, the terms “axial” and “axially” refer to directions and orientations that extend substantially parallel to a centerline of the turbine engine. Moreover, the terms “radial” and “radially” refer to directions and orientations that extend substantially perpendicular to the centerline of the turbine engine. In addition, as used herein, the terms “circumferential” and “circumferentially” refer to directions and orientations that extend arcuately about the centerline of the turbine engine.

[0024] As used herein, a “hot side” is a side of a combustion section of the turbine engine that is exposed to, or is otherwise oriented to face, a combustion chamber of the combustion section.

[0025] As used herein, a “cold side” is a side of the combustion section of the turbine engine that is not exposed to, or otherwise not oriented to face, the combustion chamber.

[0026] The present disclosure provides for a coupling assembly for a combustion section of a turbine engine. The combustion section includes a cold side component and a hot side component coupled to the cold side component. The cold side component and the hot side component can be components of a liner and/or of an annular dome assembly of the combustion section. For example,

the cold side component can be a shell of the liner and the hot side component can be a heat shield including tiles or panels coupled to the shell. In some embodiments, the cold side component is an annular dome assembly and the hot side component is a deflector assembly including tiles or panels coupled to the annular dome assembly. The hot side component protects or shields the cold side component from hot combustion gases inside a combustion chamber of the combustion section. The coupling assembly includes one or more fastening mechanisms positioned about the cold side component to fasten and to secure the hot side component to the hot side component.

[0027] When the cold side component and the hot side component are assembled, the one or more fastening mechanisms may be exposed to the hot combustion gases. For example, the one or more fastening mechanisms can include studs that are an integral part of the panels, and the studs may experience distress due to lack of adequate cooling. Accordingly, current combustion sections may experience thermal distress around the fastening mechanisms due to the proximity of the fastening mechanisms to the hot combustion gases and a lack of adequate cooling about the fastening mechanisms. In some instances, the thermal distress around the fastening mechanisms may cause fatigue, failure, or wear to a portion of the combustion section about the fastening mechanisms. Thus, embodiments of the present disclosure provide for an improved coupling assembly and cooling arrangement about the fastening mechanisms of the combustion section to improve the durability and the life cycle of such combustion sections as compared to combustion sections without the benefit of the present disclosure.

[0028] The present disclosure provides for avoiding direct exposure of hot combustion gases to the studs by providing a hollow stud and/or a hollow cap positioned on the stud. Cooling air is directed through the hollow stud and/or through the hollow cap. The present disclosure provides for various arrangements of coupling the hollow cap to the stud. The present disclosure also provides for avoiding hot combustion gas exposure by shifting the location of the head of the stud from the hot side of the panels to the shell. Embodiments of the present disclosure also include plates for retaining the panels on the cold side component without the use of rigid bolting. The plates can include a multi-finger or a projection engagement with the panels to hold the panels to the cold side component without the fastening mechanisms having to contact the panels directly. The plates can be made from a ceramic matrix composite or from a metal. Cooling air is directed through the plates. Thus, the present disclosure provides for reducing the thermal gradients and preventing stress concentration in the areas about the fastening mechanisms by facilitating cooling to the panels through the cap and/or through the plates, as compared to coupling assemblies without the benefit of the present disclosure.

[0029] Referring now to the drawings, FIG. 1 is a schematic cross-sectional diagram of a turbine engine 10, taken along a centerline axis of the turbine engine 10, according to an embodiment of the present disclosure. As shown in FIG. 1, the turbine engine 10 defines an axial direction A (extending parallel to a longitudinal, centerline axis 12 provided for reference) and a radial direction R that is normal to the axial direction A. In general, the turbine engine 10 includes a fan section 14 and a core turbine engine 16 disposed downstream from the fan section 14.

[0030] The core turbine engine 16 depicted generally includes an outer casing 18 that is substantially tubular and defines an annular inlet 20. As schematically shown in FIG. 1, the outer casing 18 encases, in serial flow relationship, a compressor section 21 including a booster or a low pressure (LP) compressor 22 followed downstream by a high pressure (HP) compressor 24, a combustion section 26, a turbine section 27 including a high pressure (HP) turbine 28 followed downstream by a low pressure (LP) turbine 30, and a jet exhaust nozzle section 32. A high pressure (HP) shaft 34 or spool drivingly connects the HP turbine 28 to the HP compressor 24 to rotate the HP turbine 28 and the HP compressor 24 in unison. A low pressure (LP) shaft 36 drivingly connects the LP turbine 30 to the LP compressor 22 to rotate the LP turbine 30 and the LP compressor 22 in unison. The compressor section 21, the combustion section 26, the turbine section 27, and the jet exhaust nozzle section 32 together define a core air flowpath.

[0031] For the embodiment depicted in FIG. 1, the fan section 14 includes a fan 38 (e.g., a variable pitch fan) having a plurality of fan blades 40 coupled to a disk 42 in a spaced apart manner. As depicted in FIG. 1, the fan blades 40 extend outwardly from the disk 42 generally along the radial direction R. Each fan blade 40 is rotatable relative to the disk 42 about a pitch axis P by virtue of the fan blades 40 being operatively coupled to an actuation member 44 configured to collectively vary the pitch of the fan blades 40 in unison. The fan blades 40, the disk 42, and the actuation member 44 are together rotatable about the centerline axis 12 via a fan shaft 45 that is powered by the LP shaft 36 across a power gearbox, also referred to as a gearbox assembly 46. The gearbox assembly 46 is shown schematically in FIG. 1. The gearbox assembly 46 includes a plurality of gears for adjusting the rotational speed of the fan shaft 45 and, thus, the fan 38 relative to the LP shaft 36.

[0032] Referring still to the exemplary embodiment of FIG. 1, the disk 42 is covered by a rotatable fan hub 48 aerodynamically contoured to promote an airflow through the plurality of fan blades 40. In addition, the fan section 14 includes an annular fan casing or a nacelle 50 that circumferentially surrounds the fan 38 and/or at least a portion of the core turbine engine 16. The nacelle 50 is supported relative to the core turbine engine 16 by a plurality of circumferentially spaced outlet guide vanes 52. Moreover, a downstream section 54 of the nacelle 50 extends over an outer portion of the core turbine engine 16 to define a bypass airflow passage 56 therebetween.

[0033] During operation of the turbine engine 10, a volume of air 58 enters the turbine engine 10 through an inlet 60 of the nacelle 50 and/or the fan section 14. As the volume of air 58 passes across the fan blades 40, a first portion of air 62 is directed or routed into the bypass airflow passage 56, and a second portion of air 64 is directed or is routed into the upstream section of the core air flowpath, or, more specifically, into the annular inlet 20 of the LP compressor 22. The ratio between the first portion of air 62 and the second portion of air 64 is commonly known as a bypass ratio. The pressure of the second portion of air 64 is then increased, forming compressed air 65, and the compressed air 65 is routed through the HP compressor 24 and into the combustion section 26, where the compressed air 65 is mixed with fuel and burned to provide combustion gases 66.

[0034] The combustion gases 66 are routed into the HP turbine 28 and expanded through the HP turbine 28 where a portion of thermal and/or of kinetic energy from the combustion gases 66 is extracted via sequential stages of HP turbine stator vanes 68 that are coupled to the outer casing 18 and HP turbine rotor blades 70 that are coupled to the HP shaft 34, thus, causing the HP shaft 34 to rotate, thereby supporting operation of the HP compressor 24. The combustion gases 66 are then routed into the LP turbine 30 and expanded through the LP turbine 30. Here, a second portion of thermal and kinetic energy is extracted from the combustion gases 66 via sequential stages of LP turbine stator vanes 72 that are coupled to the outer casing 18 and LP turbine rotor blades 74 that are coupled to the LP shaft 36, thus, causing the LP shaft 36 to rotate, thereby supporting operation of the LP compressor 22 and rotation of the fan 38 via the gearbox assembly 46.

[0035] The combustion gases 66 are subsequently routed through the jet exhaust nozzle section 32 of the core turbine engine 16 to provide propulsive thrust. Simultaneously, the pressure of the first portion of air 62 is substantially increased as the first portion of air 62 is routed through the bypass airflow passage 56 before being exhausted from a fan nozzle exhaust section 76 of the turbine engine 10, also providing propulsive thrust. The HP turbine 28, the LP turbine 30, and the jet exhaust nozzle section 32 at least partially define a hot gas path 78 for routing the combustion gases 66 through the core turbine engine 16.

[0036] The turbine engine 10 depicted in FIG. 1 is by way of example only. In other exemplary embodiments, the turbine engine 10 may have any other suitable configuration. For example, in other exemplary embodiments, the fan 38 may be configured in any other suitable manner (e.g., as a fixed pitch fan) and further may be supported using any other suitable fan frame configuration. Moreover, in other exemplary embodiments, any other suitable number or configuration of compressors, turbines, shafts, or a combination thereof may be provided. In still other exemplary

embodiments, aspects of the present disclosure may be incorporated into any other suitable turbine engine, such as, for example, turbofan engines, propfan engines, turbojet engines, turboprop, and/or turboshaft engines.

[0037] FIG. 2 is a schematic partial cross-sectional view of a portion of an exemplary combustion section 26 of the turbine engine 10 (FIG. 1), according to the present disclosure. Various embodiments of the combustion section 26 may further define a rich burn combustor in particular. Other embodiments may, however, define a lean burn combustor configuration. In the exemplary embodiment, the combustion section 26 includes an annular combustor. One skilled in the art will appreciate that the combustor may be any other combustor, including, but not limited to, a single or a double annular combustor, a can-combustor, or a can-annular combustor.

[0038] FIG. 2 shows the combustion section 26 defines an axial direction A and a radial direction R that is normal to the axial direction A. The combustion section 26 includes an outer liner 202 and an inner liner 204 disposed between an outer combustor casing 206 and an inner combustor casing 208. The outer liner 202 and the inner liner 204 are spaced radially from each other such that a combustion chamber 210 is defined therebetween. The outer liner 202 and the outer combustor casing 206 form an outer passage 212 therebetween, and the inner liner 204 and the inner combustor casing 208 form an inner passage 214 therebetween. The outer liner 202 and the inner liner 204 each includes a shell 203 and a heat shield 205. In the exemplary embodiments, the outer liner 202 and the inner liner 204 (e.g., the shell 203 and heat shield 205) are generally cylindrical, but may take any known shape of a liner for a combustor.

[0039] The heat shield 205 includes one or more tiles or panels 207 arranged on and coupled to a hot side of the shell 203. That is, the panels 207 of the heat shield 205 are coupled on a side of the shell 203 directly exposed to the combustion chamber 210. FIG. 2 depicts three panels 207 of the heat shield 205, but the heat shield 205 may include any number of panels 207, as desired. The shell 203 is made of, for example, non-ceramic materials, such as metals or the like. The panels 207 of the heat shield 205 are made of, for example, a ceramic material. In some embodiments, the panels 207 are ceramic matrix composites (CMC). Thus, the heat shield 205 provides a shield for the shell 203, enhancing the life of the shell 203, thereby enhancing the life of the outer liner 202 and of the inner liner 204.

[0040] The combustion section 26 may also include a combustor assembly 218 comprising an annular dome assembly 220 mounted upstream of the combustion chamber 210. The combustor assembly 218 is configured to be coupled to the forward ends of the outer liner 202 and the inner liner 204. More particularly, the combustor assembly 218 includes an inner annular dome 222 attached to the forward end of the inner liner 204 and an outer annular dome 224 attached to the forward end of the outer liner 202.

[0041] The combustion section 26 may be configured to receive the compressed air 65 from a discharge outlet of the HP compressor 24 (FIG. 1) of the turbine engine 10 (FIG. 1). To assist in directing the compressed air 65, the annular dome assembly 220 may further comprise an inner cowl 228 and an outer cowl 230 that may be coupled to the upstream ends of the inner liner 204 and the outer liner 202, respectively. In this regard, an annular opening 232 formed between the inner cowl 228 and the outer cowl 230 enables compressed fluid (e.g., the compressed air 65) to enter the combustion section 26 through a diffuser opening in a direction generally indicated by flow direction 234. The compressed air 65 may enter into a cavity 236 defined at least in part by the annular dome assembly 220. In various embodiments, the cavity 236 is more specifically defined between the inner annular dome 222 and the outer annular dome 224, and the inner cowl 228 and the outer cowl 230. As will be discussed in more detail below, a portion of the compressed air 65 in the cavity 236 may be used for combustion, while another portion may be used for cooling the combustion section 26.

[0042] In addition to directing air into the cavity 236 and the combustion chamber 210, the inner cowl 228 and the outer cowl 230 may direct a portion of the compressed air 65 around the outside

of the combustion chamber **210** to facilitate cooling the outer liner **202** and the inner liner **204**. For example, as shown in FIG. 2 a portion of the compressed air **65** may flow around the combustion chamber **210**, as indicated by an outer passage flow direction **238** and an inner passage flow direction **240**, to provide cooling air to the outer passage **212** and the inner passage **214**, respectively.

[0043] In certain exemplary embodiments, the inner annular dome **222** may be formed integrally as a single annular component, and, similarly, the outer annular dome **224** may also be formed integrally as a single annular component. In still certain embodiments, the inner annular dome **222** and the outer annular dome **224** may together be formed as a single integral component. In still various embodiments, the annular dome assembly **220**, including one or more of the inner annular dome **222**, the outer annular dome **224**, the outer liner **202**, or the inner liner **204**, may be formed as a single integral component. In other exemplary embodiments, the inner annular dome **222** or the outer annular dome **224** may alternatively be formed by one or more components joined in any suitable manner. For example, with reference to the outer annular dome **224**, in certain exemplary embodiments, the outer cowl **230** may be formed separately from the outer annular dome **224** and attached to the forward end of the outer annular dome **224** using, e.g., a welding process, a mechanical fastener, a bonding process or adhesive, or a composite layup process. Additionally, or alternatively, the inner annular dome **222** may have a similar configuration.

[0044] The combustor assembly **218** further includes a plurality of mixer assemblies **242** spaced along a circumferential direction between the outer annular dome **224** and the inner annular dome **222**. In this regard, the annular dome assembly **220** defines an opening in which a swirler, a cyclone, or a mixer assembly **242** is mounted, attached, or otherwise integrated for introducing the air/fuel mixture into the combustion chamber **210**. Notably, the compressed air **65** is directed from the combustion section **26** into or through one or more of the mixer assemblies **242** to support combustion in the upstream end of the combustion chamber **210**.

[0045] A liquid or a gaseous fuel is transported to the combustion section **26** by a fuel distribution system (not shown), where it is introduced at the front end of a burner in a highly atomized spray from a fuel nozzle. In an exemplary embodiment, each mixer assembly **242** defines an opening for receiving a fuel injector **246** (details are omitted for clarity). The fuel injector **246** injects fuel in a generally axial direction A, as well as in a generally radial direction R, where the fuel is swirled with the incoming compressed air **65**. Thus, each mixer assembly **242** receives compressed air **65** from the annular opening **232** and fuel from a corresponding fuel injector **246**. Fuel and pressurized air are swirled and mixed together by the mixer assemblies **242**, and the resulting fuel/air mixture is discharged into combustion chamber **210** for combustion thereof. The mixer assemblies **242** and the fuel injectors **246** may include any type of mixer assembly and fuel injector, respectively, as desired.

[0046] The combustion section **26** may further comprise an ignition assembly (e.g., one or more igniters extending through the outer liner **202**) suitable for igniting the fuel-air mixture. Details of the fuel injectors and the ignition assembly are omitted in FIG. 2 for clarity. Upon ignition, the resulting hot combustion gases may flow in a generally axial direction A through the combustion chamber **210** into and through the turbine section **27** (FIG. 1) of the turbine engine **10** (FIG. 1) where a portion of the thermal energy or the kinetic energy from the combustion gases **66** is extracted, as detailed above. More specifically, the combustion gases **66** may flow into an annular, first stage turbine nozzle **248**. As is generally understood, the first stage turbine nozzle **248** is defined by an annular flow channel that includes a plurality of radially extending, circularly spaced nozzle vanes **250** that turn the gases so that they flow angularly and impinge upon the HP turbine rotor blades **70** (FIG. 1) of the HP turbine **28** (FIG. 1).

[0047] Each dome (e.g., the inner annular dome **222** and the outer annular dome **224**) has a heat shield, for example, a deflector assembly **260**, which thermally insulates the annular dome assembly **220** from the extremely high temperatures generated in the combustion chamber **210**

during engine operation (e.g., from the combustion gases **66**). The inner annular dome **222**, the outer annular dome **224**, and the deflector assembly **260** may define a plurality of openings **244** for receiving the mixer assemblies **242**. As shown, the plurality of openings **244** are, in one embodiment, circular. In other embodiments, the plurality of openings **244** are ovalar, elliptical, polygonal, oblong, or other non-circular cross sections. The deflector assembly **260** is mounted on a combustion chamber side (e.g., a hot side) of the annular dome assembly **220**.

[0048] The deflector assembly **260** includes one or more tiles or panels **261** arranged on and coupled to a hot side of the annular dome assembly **220**. That is, the panels **261** of the deflector assembly **260** are coupled on a side of the annular dome assembly **220** directly exposed to the combustion chamber **210**. FIG. 2 depicts two panels **261** of the deflector assembly **260**, but the deflector assembly **260** may include any number of panels **261**, as desired. The panels **261** of the deflector assembly **260** are made of, for example, a ceramic material. In some embodiments, the panels **261** are ceramic matrix composites (CMC). Thus, the deflector assembly **260** provides a shield for the annular dome assembly **220**, thereby enhancing the life of the annular dome assembly **220**.

[0049] FIGS. 3 to 9B depict various embodiments of a coupling assembly for a combustion section **26** (FIGS. 1 and 2) of a turbine engine **10** (FIG. 1), taken at a centerline axis of the turbine engine **10** (FIG. 1). The coupling assemblies of FIGS. 3 to 9B can be used for the outer liner **202** (FIG. 2), the inner liner **204** (FIG. 2), the annular dome assembly **220** (FIG. 2), and/or any other component of the combustion section **26** or of the turbine engine **10**. Each of the coupling assemblies detailed herein includes coupling a hot side component to a cold side component. As used herein, the “hot side” is a side of the combustion section **26** (FIG. 2) that is exposed to, or is otherwise oriented to face, the combustion chamber **210**, and the “cold side” is a side of the combustion section **26** (FIG. 2) that is not exposed to, or is not otherwise oriented to face, the combustion chamber **210**. In the exemplary embodiments, the cold side component is the shell **203** (FIG. 2) of the outer liner **202** (FIG. 2) or of the inner liner **204** (FIG. 2), and the hot side component is the heat shield **205** (FIG. 2). In some embodiments, the cold side component is the inner annular dome **222** (FIG. 2) and/or the outer annular dome **224** (FIG. 2), and the hot side component is the deflector assembly **260** (FIG. 2). FIGS. 3 to 9B include many of the same or similar components and functionality. Similar reference numerals denote similar features and functionality between the figures, unless noted otherwise. The embodiments of FIGS. 3 to 9B are detailed further below.

[0050] FIG. 3 is a schematic partial cross-sectional view of a portion of a coupling assembly **300** for a combustion section (FIGS. 1 and 2) of a turbine engine **10** (FIG. 1), taken at a centerline axis of the turbine engine **10** (FIG. 1), according to the present disclosure. FIG. 3 shows the coupling assembly **300** includes a cold side component **302** and a hot side component **304**. The hot side component **304** includes one or more tiles or panels **306** arranged on and coupled to a hot side of the cold side component **302**. That is, the panels **306** of the hot side component **304** are coupled on a side of the cold side component **302** exposed to the combustion chamber **210**. One panel **306** is depicted in FIG. 3, but the hot side component **304** may include any number of panels **306**, as desired.

[0051] The cold side component **302** includes one or more cold side component cooling holes **308** and the hot side component **304** includes one or more hot side component cooling holes **310**. In this way, cooling air **380** (e.g., the compressed air **65** in FIG. 2) can pass through the cold side component **302** and can pass through the hot side component **304**, as detailed further below. The one or more cold side component cooling holes **308** are impingement cooling holes that are disposed generally radially through the cold side component **302**. The one or more hot side component cooling holes **310** are film cooling holes that are disposed through the hot side component **304** at an angle. The one or more cold side component cooling holes **308** and the one or more hot side component cooling holes **310** may include any number of cooling holes formed in any orientation with respect to the cold side component **302** and the hot side component **304**,

respectively. In some embodiments, the one or more cold side component cooling holes **308** and the one or more hot side component cooling holes **310** are also angled circumferentially to operably direct cooling air tangentially.

[0052] The coupling assembly **300** includes one or more fastening mechanisms **320** for attaching and connecting the cold side component **302** and the hot side component **304** (e.g., each of the panels **306** of the hot side component **304**). That is, the hot side component **304** is coupled by the one or more fastening mechanisms **320** to the cold side component **302**. While FIG. **3** shows one fastening mechanism **320**, the one or more fastening mechanisms **320** may include any number of fastening mechanisms, as desired. A gap or a space **322** is formed between the cold side component **302** and the hot side component **304**. For example, the space **322** is formed between a radially outer surface of hot side component **304** and a radially inner surface of the cold side component **302**.

[0053] The one or more fastening mechanisms **320** include a stud **324** and a cap **326**. The one or more fastening mechanisms **320** may include any type of known fastening mechanism such as, for example, bolts, screws, nuts, rivets, or the like. The stud **324** includes a generally cylindrical body that defines the stud **324**. The cap **326** includes a generally frustoconical shape with a hollow interior **328** such that the cap **326** includes an exterior surface **330** and an interior surface **332**. The exterior surface **330** and the interior surface **332** together form a thin wall **333** such that the hollow interior **328** includes a shape that generally corresponds to a shape of the cap **326**. In this way, the hollow interior **328** is an enclosed hollow interior **328**. Accordingly, the hollow interior **328** defines a reservoir that is filled with cooling air during operation of the turbine engine **10** (FIG. **1**), as detailed further below. The stud **324** and the cap **326** may include any shape and/or any size, as desired. The cap **326** and the hollow interior **328** each include a diameter larger than an outer diameter of the stud **324**. The stud **324** is inserted through a first fastening mechanism aperture **334** of the cold side component **302** and the cap **326** is coupled to the stud **324**. When the cap **326** is coupled to the stud **324**, the cap **326** is inserted and is disposed in a second fastening mechanism aperture **336** of the hot side component **304**. In this way, the cap **326** seals the second fastening mechanism aperture **336** such that the hot combustion gases in the combustion chamber **210** do not leak through the second fastening mechanism aperture **336**. The cap **326** is positioned between the stud **324** and the combustion chamber **210** such that the stud **324** is insulated from the combustion chamber **210**. In this way, the stud **324** is not directly exposed to the combustion chamber **210**. For example, the stud **324** is spaced radially from the combustion chamber **210**.

[0054] The cap **326** is coupled to the stud **324** by a first threaded connection **338**. In some embodiments, the cap **326** is press fit onto the stud **324** without the use of a threaded connection. In some embodiments, the cap **326** is formed integral with the stud **324** such that the cap **326** and the stud **324** form a single, unitary component. The one or more fastening mechanisms **320** also include a nut **340** that is coupled to the stud **324** on a cold side of the stud **324**. The nut **340** is coupled to the stud **324** by a second threaded connection **342**. In some examples, the nut **340** is press fit to the stud **324**. As the nut **340** is tightened, the second threaded connection **342** pulls the stud **324** and the cap **326** through the second fastening mechanism aperture **336** such that the fastening mechanism **320** is tightened and secured to securely couple the hot side component **304** to the cold side component **302**.

[0055] The cap **326** includes a stud receiving portion **344** that defines a stud opening **346** for receiving a portion of the stud **324**. The stud receiving portion **344** and the stud opening **346** are sized and are shaped to receive the stud **324**. FIG. **3** shows the stud receiving portion **344** includes a solid, annular body about the stud **324**. In some embodiments, the cap **326** can define a radial gap or a radial space about the stud receiving portion **344** for additional cooling, as detailed further below with respect to FIG. **5**. The stud receiving portion **344** extends from a cold side of the interior surface **332** and into the hollow interior **328** of the cap **326**. The stud receiving portion **344** is spaced from the hot side of the interior surface **332**. In this way, the hollow interior **328** surrounds the stud receiving portion **344**.

[0056] The cap **326** can include a cap insert **348** that is threaded onto the stud **324** at the first threaded connection **338**. The stud opening **346** is sized and is shaped to receive the cap insert **348** in order to couple the cap **326** to the stud **324**. For example, the cap **326** is press fit onto the cap insert **348**. In some embodiments, the cap insert **348** is formed integrally with the cap **326** such that the cap insert **348** and the cap **326** form a single, unitary component. In some embodiments, the cap insert **348** is welded, brazed, or otherwise attached to the cap **326** and/or to the stud **324**. In some embodiments, the cap insert **348** is press fit onto the stud **324**. In some embodiments, the cap insert **348**, the cap **326**, and the stud **324** are formed integrally together such that the cap insert **348**, the cap **326**, and the stud **324** form a single, unitary component. In some embodiments, the cap **326** is directly coupled to the stud **324** without the use of a cap insert. For example, the stud receiving portion **344** and the stud opening **346** of the cap **326** can be sized and shaped to directly receive the stud **324**. In some embodiments, the stud receiving portion **344** includes threads such that the cap **326** is threaded to couple the cap **326** to the stud **324** at the first threaded connection **338** by the threads of the stud receiving portion **344**.

[0057] The cap **326** includes one or more first cap cooling holes **350** extending therethrough, and one or more second cap cooling holes **352** extending therethrough. The one or more first cap cooling holes **350** and the one or more second cap cooling holes **352** each extends through the thin wall **333** from the exterior surface **330** to the interior surface **332** of the cap **326**. The one or more first cap cooling holes **350** are disposed in the space **322** when the one or more fastening mechanisms **320** are secured. In this way, the one or more first cap cooling holes **350** provide fluid communication from the space **322** to the hollow interior **328** of the cap **326**. The one or more second cap cooling holes **352** are disposed at a hot side of the cap **326**. In this way, the one or more second cap cooling holes **352** provide fluid communication from the hollow interior **328** of the cap **326** to the combustion chamber **210**. In some embodiments, the one or more first cap cooling holes **350** and/or the one or more second cap cooling holes **352** are angled circumferentially to operably direct cooling air tangentially to the hollow interior **328** or to the combustion chamber **210**, respectively.

[0058] The coupling assembly **300** also includes a washer **360** that is disposed in the space **322** between the cold side component **302** and the hot side component **304**. The washer **360** includes a first washer surface **362** and a second washer surface **364** opposite the first washer surface **362**. The washer **360** includes a flexible washer such that the washer **360** can bend or otherwise can flex with movement of the cold side component **302** and/or of the hot side component **304**. The washer **360** includes a flat portion **366** and a bend portion **368**. The flat portion **366** includes a generally straight, flat portion of the washer **360**. The bend portion **368** includes one or more bends and defines a generally frustoconical shape that surrounds the portion of the cap **326** within the space **322**.

[0059] When the coupling assembly **300** is assembled, a cold side of the exterior surface **330** of the cap **326** contacts the second washer surface **364** at the flat portion **366** of the washer **360** such that the washer **360** is disposed and secured between the cold side component **302** and the cap **326**. The bend portion **368** is bent from the flat portion **366** such that the bend portion **368** extends to the cold side of the hot side component **304**. In this way, the first washer surface **362** contacts the cold side component **302** and the second washer surface **364** contacts the cap **326** (e.g., at the flat portion **366**) and contacts the hot side component **304** (e.g., at the bend portion **368**). Thus, the washer **360** maintains the space **322** between the cold side component **302** and the hot side component **304** such that the washer **360** prevents the hot side component **304** from contacting the cold side component **302** during operation of the combustion section **26** (FIG. 2).

[0060] The washer **360** includes a washer stud hole **370** for receiving the stud **324** therethrough. In this way, the washer **360** is secured by the fastening mechanism **320** when the fastening mechanism **320** is assembled and fastened as detailed above. The washer **360** includes one or more washer cooling holes **372** that each extends through the washer **360** from the first washer surface **362** to the

second washer surface **364**. The one or more washer cooling holes **372** are disposed in the bend portion **368** of the washer **360**. The one or more washer cooling holes **372** are disposed within the space **322** when the coupling assembly **300** is assembled. In this way, the cooling air **380** is operably directed from the space **322** through the washer **360** and between the washer **360** and the cap **326**, as detailed further below. The one or more washer cooling holes **372** may include any number of cooling holes, as desired.

[0061] In operation, cooling air **380** is caused to operably flow from the cold side of the cold side component **302**, through the one or more cold side component cooling holes **308**, and into the space **322**. In the exemplary embodiment, the cooling air **380** is the compressed air **65** (FIG. 2) received from the compressor section of the turbine engine. The cooling air **380**, however, may be supplied from any source, as desired. The cooling air **380** is operably directed from the space **322**, through the one or more washer cooling holes **372**, and between the washer **360** and the cap **326**. At the same time, the cooling air **380** is operably directed from the space **322** through the one or more hot side component cooling holes **310** and into the combustion chamber **210** to film cool the hot side component **304**. The cooling air **380** is operably directed from between the washer **360** and the cap **326** through the one or more first cap cooling holes **350** and into the hollow interior **328** of the cap **326**. In this way, the cooling air **380** fills the hollow interior **328**. The cooling air **380** is operably directed from the hollow interior **328** through the one or more second cap cooling holes **352** and into the combustion chamber **210** to film cool the cap **326**.

[0062] Accordingly, the cap **326** having a hollow interior **328**, the one or more first cap cooling holes **350**, and the one or more second cap cooling holes **352** provides for a cooling arrangement that avoids direct exposure of the stud **324** to the hot combustion gases in the combustion chamber **210**. The cooling air **380** in the hollow interior **328** provides a cushion of cooling air **380** between the combustion chamber **210** and the stud **324**. The film cooling through the cap **326** provides a layer of cooling air **380** between the hot side of the exterior surface **330** of the cap **326** and the hot air stream (e.g., the hot combustion gases). The film cooling thus protects the cap **326**, and the cushion of cooling air **380** in the hollow interior **328** of the cap **326** protects the stud **324** from being exposed to the hot combustion gases and provides additional cooling for the stud **324**. Therefore, the thermal gradients in an area of one or more fastening mechanisms **320** are reduced, thereby reducing the stress concentration on the one or more fastening mechanisms **320**. Thus, the coupling assembly **300** provides for reducing damage to and/or for reducing failure of the hot side component **304** and/or of the cold side component **302** in an area about the one or more fastening mechanisms **320**.

[0063] FIG. 4 is a top view of the washer **360** isolated from the coupling assembly **300** (FIG. 3), according to the present disclosure. FIG. 4 shows the washer **360** in an un-bent orientation such that washer **360** is flattened. The washer **360** can be bent or otherwise can be distorted to form the generally frustoconical shape shown in FIG. 3. In the un-bent configuration, the washer **360** defines a generally circular shape. The washer **360** may include any shape and/or any size, as desired. The washer **360** includes an outer edge **474** that defines an outer perimeter of the washer **360**. The washer stud hole **370** is located at a radial center of the washer **360**. The one or more washer cooling holes **372** are disposed radially outward from the washer stud hole **370** and are disposed circumferentially about the washer **360**. In this way, the one or more washer cooling holes **372** are located between the outer edge **474** and the washer stud hole **370**.

[0064] The washer **360** also includes one or more washer slots **476** disposed circumferentially about the outer edge **474**. Each washer slot **476** extends radially inward from the outer edge **474** to a respective washer cooling hole **372**. In this way, the one or more washer slots **476** allow the washer **360** to be bent in an area about the outer edge **474** such that the washer **360** is fit between the cold side component **302** (FIG. 3) and the hot side component **304** (FIG. 3). The bending of the washer **360** allows the washer **360** to flex or to compress such that the hot side component **304** can move radially with respect to the cold side component **302** and prevents the hot side component

304 from contacting the cold side component **302**.

[0065] FIG. 5 is a schematic partial cross-sectional view of a portion of a coupling assembly **500** for a combustion section (FIGS. 1 and 2) of a turbine engine **10** (FIG. 1), taken at a centerline axis of the turbine engine **10** (FIG. 1), according to another embodiment. The coupling assembly **500** is substantially similar to the coupling assembly **300** of FIG. 3. For example, the coupling assembly **500** includes one or more fastening mechanisms **520** including the stud **324** and a cap **526**. The cap **526** includes a generally frustoconical shape with a hollow interior **528** such that the cap **526** includes an exterior surface **530** and an interior surface **532**. The exterior surface **530** and the interior surface **532** together form a thin wall **533** such that the hollow interior **528** includes a shape that generally corresponds to a shape of the cap **526**. In this way, the hollow interior **528** is an enclosed hollow interior **528**. Accordingly, the hollow interior **528** defines a reservoir that is filled with cooling air during operation of the turbine engine **10** (FIG. 1), as detailed further below. The cap **526** and the hollow interior **528** each include a diameter larger than an outer diameter of the stud **324**. The cap **526** is positioned between the stud **324** and the combustion chamber **210** such that the stud **324** is insulated from the combustion chamber **210**. In this way, the stud **324** is not directly exposed to the combustion chamber **210**. For example, the stud **324** is spaced radially from the combustion chamber **210**. The cap **526** also includes one or more first cap cooling holes **550** and one or more second cap cooling holes **552**. The cap **526**, however, is different than the cap **326** of FIG. 3.

[0066] The cap **526** includes a stud receiving portion **544** that defines a stud opening **546** for receiving a portion of the stud **324**. The cap **526** defines a radial gap or a radial space **547** about the stud receiving portion **544**. The radial space **547** extends radially from the exterior surface **530** of the stud receiving portion **544** to another portion of the exterior surface **530** of the cap **526**. For example, the exterior surface **530** defines a generally U-shaped portion in an area of the radial space **547**. The radial space **547** may include any size and/or any shape, as desired. The stud receiving portion **544** abuts the washer **360** when the coupling assembly **500** is assembled. The cap **526** also includes one or more third cap cooling holes **553** in the generally U-shaped portion of the exterior surface **530**. A portion of the cold side of the exterior surface **530** of the cap **526** is spaced from the washer **360**.

[0067] In operation, the cooling air **380** is operably directed between the washer **360** and the cold side of the exterior surface **530** of the cap **526**, and into the radial space **547** to cool the stud receiving portion **544**, thereby cooling the stud **324**. The cooling air **380** is operably directed from the radial space **547** through the one or more third cap cooling holes **553** and into the hollow interior **528**. In this way, the cooling air **380** fills the hollow interior **528** to provide a cushion of cooling air **380** between the combustion chamber **210** and the stud **324**. The cooling air **380** is operably directed from the hollow interior **528** through the one or more second cap cooling holes **552** and into the combustion chamber **210** to provide film cooling, as detailed above with respect to FIG. 3. Accordingly, the radial space **547** provides additional cooling about the stud receiving portion **544**, thereby providing additional cooling about the stud **324**. The radial space **547** also provides additional flexibility such that the cap **526** can flex or can compress in an area about the radial space **547**. Thus, if the hot combustion gases cause the cap **526** to move, the stress and the strain onto the stud **324** from the movement of the cap **526** is reduced.

[0068] FIG. 6 is a schematic partial cross-sectional view of a portion of a coupling assembly **600** for a combustion section **26** (FIGS. 1 and 2) of a turbine engine **10** (FIG. 1), taken at a centerline axis of the turbine engine **10** (FIG. 1), according to another embodiment. The coupling assembly **600** includes the cold side component **302** and a hot side component **604**. The hot side component **604** includes one or more tiles or panels **606** arranged on and coupled to a hot side of the cold side component **302**. One panel **606** is depicted in FIG. 6, but the hot side component **604** may include any number of panels **606**, as desired. The coupling assembly **600** includes one or more fastening mechanisms **620** for attaching and for connecting the hot side component **604** to the cold side

component **302**. The hot side component **604** includes one or more hot side component cooling holes **610**.

[0069] Each panel **606** of the hot side component **604** includes a recessed portion **612** in an area about each of the one or more fastening mechanisms **620**. The recessed portion **612** includes a portion of a respective panel **606** that is recessed with respect to the hot side of the hot side component **604**. For example, the recessed portion **612** may extend from the hot side component **604** towards the cold side component **302**. The recessed portion **612** may be the same or may be substantially similar to the recessed portion described in U.S. application Ser. No. 17/812,897, the contents of which are herein incorporated by reference in their entirety. A respective fastening mechanism **620** may be located centrally in the recessed portion **612** such that the respective fastening mechanism **620** is recessed or is otherwise retracted with respect to the hot side of the hot side component **604**. The respective fastening mechanism **620** may be located anywhere within the recessed portion **612**, as desired. Further, while the recessed portion **612** in the exemplary embodiment is generally circular in shape, the recessed portion **612** about each fastening mechanism **620** may include any shape or any size, as desired. The washer **360** is coupled between the hot side of the cold side component **302** and the cold side of the recessed portion **612**. The washer **360** includes a larger diameter than that of the recessed portion **612** such that the washer **360** circumferentially surrounds the recessed portion **612**. The recessed portion **612** includes one or more recessed portion cooling holes **614** extending therethrough. The one or more recessed portion cooling holes **614** extend from the cold side of the recessed portion **612** to the hot side of the recessed portion **612**.

[0070] The recessed portion **612** also includes a stud receiving portion **616** that defines a gap or a space that is sized and shaped to receive a stud **624** of a respective fastening mechanism **620**. The one or more fastening mechanisms **620** include the stud **624** and a cap **626**. The stud **624** includes one or more stud cooling holes **625** extending therethrough. The one or more stud cooling holes **625** extend longitudinally through a length of the stud **624** from a cold side of the stud **624** to a hot side of the stud **624**. In this way, the one or more stud cooling holes **625** provide fluid communication from the cold side of the cold side component **302** to a hollow interior **628** of the cap **626**, as detailed further below. The stud **624** also includes a cap receiving portion **627** that is sized and is shaped to receive a portion of the cap **626**, as detailed below.

[0071] The cap **626** includes a generally frustoconical shape with hollow interior **628** such that the cap **626** includes an exterior surface **630** and an interior surface **632**. The exterior surface **630** and the interior surface **632** together form a thin wall **633** such that the hollow interior **628** includes a shape that generally corresponds to a shape of the cap **626**. In this way, the hollow interior **628** is an enclosed hollow interior **628**. Accordingly, the hollow interior **628** defines a reservoir that is filled with cooling air during operation of the turbine engine **10** (FIG. **1**), as detailed further below. The stud **624** and the cap **626** may include any shape and/or any size, as desired. The cap **626** and the hollow interior **628** each include a diameter larger than an outer diameter of the stud **624**. The cap **626** is coupled to the stud **624**. When the cap **626** is coupled to the stud **624**, the cap **626** is inserted into and is disposed in a recessed portion space **636** defined by the recessed portion **612** of the hot side component **604**. A portion of the cap **626** is spaced from the cold side of the recessed portion **612** such that a gap or a space **647** is defined between the cap **626** and the recessed portion **612**. The cap **626** is positioned between the stud **624** and the combustion chamber **210** such that the stud **624** is insulated from the combustion chamber **210**. In this way, the stud **624** is not directly exposed to the combustion chamber **210**. For example, the stud **624** is spaced radially from the combustion chamber **210**.

[0072] The cap **626** includes a cap extension **644** that is sized and is shaped to fit into the cap receiving portion **627** of the stud **624**. The cap extension **644** extends from a cold side of the exterior surface **630** of the cap **626**. The cap extension **644** is inserted through an aperture in the recessed portion **612** and is received by the cap receiving portion **627** of the stud **624**. When

assembled, the cold side of the exterior surface **630** of the cap **626** abuts the hot side of the recessed portion **612**. The hot side of the exterior surface **630** of the cap **626** is substantially flush or substantially in line with the hot side of the hot side component **604**. In some embodiments, the cap **626** may extend beyond (e.g., radially inward) the hot side of the hot side component **604**. In some embodiments, the cap **626** extends to a location radially outward of the hot side of the hot side component **604**.

[0073] The recessed portion **612** includes an insert **648** disposed within the stud receiving portion **616** for coupling the stud **624** and the cap **626**. The stud **624** includes threads such that the stud **624** is coupled to the insert **648** by corresponding threads on the insert **648** at a first threaded connection **638**. The stud **624** also includes threads such that the stud **624** is coupled to a nut **640** by corresponding threads on the nut **340** at a second threaded connection **642**. The cap **626** includes threads such that the cap **626** is coupled to the insert **648** by corresponding threads on the insert **648** at a third threaded connection **645**. For example, the cap **626** is coupled to the insert **648** by threads on the cap extension **644**. In some embodiments, the stud **624** and/or the cap **626** are press fit into the insert **648** without the use of threaded connections. Any of the threaded connections may include other types of connections, such as, for example, press fitting, welding, brazing, adhesive, or the like. In some embodiments, the coupling assembly **600** does not include an insert **648** and the cap **626** is directly coupled to the stud **624**.

[0074] The cap **626** includes one or more first cap cooling holes **650** extending therethrough, one or more second cap cooling holes **652** extending therethrough, and one or more third cap cooling holes **653** extending therethrough. The one or more first cap cooling holes **650**, the one or more second cap cooling holes **652**, and the one or more third cap cooling holes **653** each extends through the thin wall **633** from the exterior surface **630** to the interior surface **632** of the cap **626**. The one or more first cap cooling holes **650** are disposed in the space **647** when the one or more fastening mechanisms **620** are secured. In this way, the one or more first cap cooling holes **650** provide fluid communication from the space **647** to the hollow interior **628** of the cap **626**. The one or more second cap cooling holes **652** are disposed at a hot side of the cap **626**. In this way, the one or more second cap cooling holes **652** provide fluid communication from the hollow interior **628** of the cap **626** to the combustion chamber **210**. The one or more third cap cooling holes **653** extend through the cap extension **644** and are aligned with the one or more stud cooling holes **625**. In this way, the one or more third cap cooling holes **653** provide fluid communication from the one or more stud cooling holes **625** to the hollow interior **628** of the cap **626**. In some embodiments, the one or more first cap cooling holes **650**, the one or more second cap cooling holes **652**, and/or the one or more third cap cooling holes **653** are angled circumferentially to operably direct cooling air tangentially to the hollow interior **628** or to the combustion chamber **210**, respectively.

[0075] In operation, in addition to the cooling arrangement detailed with respect to FIG. 3, the cooling air **380** is operably directed through the one or more washer cooling holes **372** to a space between the washer **360** and the recessed portion **612**. The cooling air **380** is operably directed from the space between the washer **360** and the recessed portion **612** through the one or more recessed portion cooling holes **614** and into the space **647** between the recessed portion **612** and the cap **626**. The cooling air **380** is operably directed from the space **647** through the one or more first cap cooling holes **650** and into the hollow interior **628** of the cap **626**. In this way, the cooling air **380** fills the hollow interior **628**. The cooling air **380** is then operably directed from the hollow interior **628** through the one or more second cap cooling holes **652** and into the combustion chamber **210** to provide film cooling at the hot side of the cap **626**. At the same time, the cooling air **380** is operably directed through the one or more stud cooling holes **625**, through the one or more third cap cooling holes **653**, into the hollow interior **628**, through the one or more second cap cooling holes **652**, and into the combustion chamber **210**. At the same time, the cooling air **380** is directed from the space **647** to the combustion chamber **210** to provide additional cooling in an area about the cap **626**.

[0076] The coupling assembly **600** provides for an improved cooling arrangement for reducing the thermal gradient in an area about the one or more fastening mechanisms **620**, as detailed above. For example, providing the recessed portion **612** moves the stud **624** away from the hot side of the hot side component **604** such that the stud **624** is spaced radially from the combustion chamber **210**, and the cap **626** provides the improved cooling detailed above with respect to FIG. 3. For example, the cooling air **380** in the hollow interior **628** provides a cushion of cooling air **380** between the combustion chamber **210** and the stud **324**. The cap **626** also provides for filling an abrupt step in the hot side component **604** (e.g., filling the recessed portion space **636**) that is formed by the recessed portion **612**. In this way, the cap **626** provides for a surface that is flush with the hot side of the hot side component **604** such that hot side of the hot side component **604** remains substantially flush and substantially smooth, thereby providing an improved cooling arrangement in an area about the one or more fastening mechanisms **620**.

[0077] FIG. 7 is a schematic partial cross-sectional view of a portion of a coupling assembly **700** for a combustion section **26** (FIGS. 1 and 2) of a turbine engine **10** (FIG. 1), taken at a centerline axis of the turbine engine **10** (FIG. 1), according to another embodiment. The coupling assembly **700** includes the cold side component **302** and a hot side component **704**. The hot side component **704** includes one or more tiles or panels **706** arranged on and coupled to a hot side of the cold side component **302**. One panel **706** is depicted in FIG. 7, but the hot side component **704** may include any number of panels **706**, as desired. The coupling assembly **700** includes one or more fastening mechanisms **720** for attaching and for connecting the hot side component **704** to the cold side component **302**. The hot side component **704** includes one or more hot side component cooling holes **710**.

[0078] The hot side component **704** is substantially similar to the hot side component **604** of FIG. 6. For example, the hot side component **704** includes a recessed portion **712** in an area surrounding each of the one or more fastening mechanisms **720**. As shown in FIG. 7, the washer **360** does not contact the recessed portion **712**. The washer **360** includes a larger diameter than that of the recessed portion **712** such that the washer **360** circumferentially surrounds the recessed portion **712**. The recessed portion **712** includes one or more recessed portion cooling holes **714** extending therethrough. The one or more recessed portion cooling holes **714** extend from the cold side of the recessed portion **712** to the hot side of the recessed portion **712**. The recessed portion **712** also includes a stud receiving portion **716** that defines a gap or a space that is sized and that is shaped to receive a stud **724** of a respective fastening mechanism **720**.

[0079] The one or more fastening mechanisms **720** include the stud **724** and a cap **726**. Although not shown, the one or more fastening mechanism **720** can also include a nut, as detailed above. The stud **624** includes a stud head **725** that is disposed in the stud receiving portion **716** of the recessed portion **712**. The stud head **725** extends from the cold side of the recessed portion **612** to the hot side of the recessed portion **612** such that the stud head **725** is flush with both the cold side and the hot side of the recessed portion **612**. The cap **726** is positioned between the stud **724** and the combustion chamber **210** such that the stud **724** is insulated from the combustion chamber **210**. In this way, the stud **724** is not directly exposed to the combustion chamber **210**. For example, the stud **724** is spaced radially from the combustion chamber **210**.

[0080] The cap **726** includes a generally frustoconical shape with a hollow interior **728** such that the cap **726** includes an exterior surface **730** and an interior surface **732**. The exterior surface **730** and the interior surface **732** together form a thin wall **733** such that the hollow interior **728** includes a shape that generally corresponds to a shape of the cap **726**. In this way, the hollow interior **728** is an enclosed hollow interior **728**. Accordingly, the hollow interior **728** defines a reservoir that is filled with cooling air during operation of the turbine engine **10** (FIG. 1), as detailed further below. The stud **724** and the cap **726** may include any shape and/or any size, as desired. The cap **726** and the hollow interior **728** each include a diameter larger than an outer diameter of the stud **724**. The cap **726** is coupled to the stud **724**. For example, a cold side of the cap **726** is coupled to the stud

head **725**. The cap **726** can be coupled to the stud **724**, for example, by welding, brazing, adhesive, or the like. In some embodiments, the cap **726** is formed integrally with the stud **724** such that the stud **724** and the cap **726** form a single, unitary component. When the cap **726** is coupled to the stud **724**, the cap **726** is inserted and is disposed in a second fastening mechanism aperture **736** defined by the recessed portion **712** of the hot side component **704**. A portion of the cap **726** is spaced from the cold side of the recessed portion **712** such that a gap or a space **747** is defined between the cap **726** and the recessed portion **712**.

[0081] The cap **726** includes one or more first cap cooling holes **750** extending therethrough and one or more second cap cooling holes **752** extending therethrough. The one or more first cap cooling holes **750** and the one or more second cap cooling holes **752** each extends through the thin wall **733** from the exterior surface **730** to the interior surface **732** of the cap **726**. The one or more first cap cooling holes **750** are disposed in the space **747** when the one or more fastening mechanisms **720** are secured. In this way, the one or more first cap cooling holes **750** provide fluid communication from the space **747** to the hollow interior **728** of the cap **726**. The one or more second cap cooling holes **752** are disposed at a hot side of the cap **726**. In this way, the one or more second cap cooling holes **752** provide fluid communication from the hollow interior **728** of the cap **726** to the combustion chamber **210**. In some embodiments, the one or more first cap cooling holes **750** and/or the one or more second cap cooling holes **752** are angled circumferentially to operably direct cooling air tangentially to the hollow interior **728** or to the combustion chamber **210**, respectively. The cooling air **380** is operably directed through the coupling assembly **700** similarly as to the coupling assembly **600** described with respect to FIG. 6. For example, the cooling air **380** is operably directed into the hollow interior **728** of the cap **726** to provide a cushion of cooling air **380** between the combustion chamber **210** and the stud **324**.

[0082] FIG. 8A is a schematic partial cross-sectional view of a portion of a coupling assembly **800** for a combustion section **26** (FIGS. 1 and 2) of a turbine engine **10** (FIG. 1), taken at a centerline axis of the turbine engine **10** (FIG. 1), according to another embodiment. FIG. 8A shows the coupling assembly **800** includes a cold side component **802** and a hot side component **804**. The hot side component **804** includes one or more tiles or panels **806** arranged on and coupled to a hot side of the cold side component **802**. That is, the panels **806** of the hot side component **804** are coupled on a side of the cold side component **802** exposed to the combustion chamber **210**. One panel **806** is depicted in FIG. 8A, but the hot side component **804** may include any number of panels **806**, as desired.

[0083] The cold side component **802** includes one or more cold side component cooling holes **808** and the hot side component **804** includes one or more hot side component cooling holes **810**. In this way, cooling air **380** (e.g., the compressed air **65** shown in FIG. 2) can pass through the cold side component **802** and can pass through the hot side component **804**, as detailed further below. The one or more cold side component cooling holes **808** and the one or more hot side component cooling holes **810** may include any number of cooling holes formed in any orientation with respect to the cold side component **802** and the hot side component **804**, respectively. In some embodiments, the one or more cold side component cooling holes **808** and the one or more hot side component cooling holes **810** are also angled circumferentially to operably direct cooling air tangentially.

[0084] The coupling assembly **800** includes a plurality of fastening mechanisms **820** for attaching and connecting the cold side component **802** and the hot side component **804** (e.g., each of the panels **806** of the hot side component **804**). That is, the hot side component **804** is coupled by the plurality of fastening mechanisms **820** to the cold side component **802**. FIG. 8A shows the plurality of fastening mechanisms **820** include a first fastening mechanism **820a** and a second fastening mechanism **820b**. The second fastening mechanism **820b** is positioned axially downstream from the first fastening mechanism **820a**. While FIG. 8A shows two fastening mechanisms **820**, the plurality of fastening mechanisms **820** may include any number of fastening mechanisms, as

desired. A gap or a space **822** is formed between the cold side component **802** and the hot side component **804**. For example, the space **822** is formed between a radially outer surface (e.g., the cold side) of hot side component **804** and a radially inner surface (e.g., the hot side) of the cold side component **802**.

[0085] The plurality of fastening mechanisms **820** each includes a stud **824** and a nut **840** connected together by a threaded connection. The plurality of fastening mechanisms **820** may include any type of known fastening mechanism such as, for example, bolts, screws, nuts, rivets, or the like. The stud **824** includes a generally cylindrical body that defines the stud **824**. The stud **824** may include any shape and/or any size, as desired. The stud **824** is inserted through a fastening mechanism aperture **834** of the cold side component **802**. For example, the stud **824** of the first fastening mechanism **820a** is inserted through a first fastening mechanism aperture **834a** and the stud **824** of the second fastening mechanism **820b** is inserted through a second fastening mechanism aperture **834b**. The stud **824** also includes a stud head **825** disposed on the hot side of the cold side component **802**. In this way, the stud head **825** is located away from the hot side component **804** such that the stud **824** and the stud head **825** are insulated from the hot combustion gases in the combustion chamber **210**. In FIG. **8A**, the stud **824** does not directly couple the hot side component **804** and the cold side component **802**. Rather, the coupling assembly **800** includes a plurality of flexible plates **850** that couple the hot side component **804** to the cold side component **802**, as detailed further below. The plurality of flexible plates **850** can be made of metal, ceramic, or combinations thereof. In some embodiments, the plurality of flexible plates **850** are made of a CMC.

[0086] The hot side component **804** includes a plurality of flanges **812** that the plurality of flexible plates **850** engage or otherwise hold to couple the hot side component **804** to the cold side component **802**. The plurality of flanges **812** include a first flange **812a** and a second flange **812b**. The second flange **812b** is positioned downstream of the first flange **812a**. In other words, the first flange **812a** is positioned upstream of the second flange **812b**.

[0087] Each of the plurality of flanges **812** includes a radial flange portion **814** that extends substantially radially from the hot side component **804**. For example, the radial flange portion **814** extends radially from a cold side of the hot side component **804** towards the cold side component **802**. The first flange **812a** includes a first radial flange portion **814a** and the second flange **812b** includes a second radial flange portion **814b**. The radial flange portion **814** of each flange **812** defines an axial end of a respective panel **806** of the hot side component **804**. For example, the first radial flange portion **814a** is positioned at a first axial end of the panel **806** and the second radial flange portion **814b** is positioned at a second axial end of the panel **806** opposite the first axial end. Each radial flange portion **814** is generally axially aligned with a respective fastening mechanism **820** such that the panel **806** extends axially between the first fastening mechanism **820a** and the second fastening mechanism **820b**. Such an arrangement provides for balancing the panel **806** when the panel **806** is coupled to the cold side component **802**, as detailed further below. The panel **806**, however, may include any axial length as desired.

[0088] Each of the plurality of flanges **812** also includes an axial flange portion **816** that extends axially from the radial flange portion **814**. The first flange **812a** includes a first axial flange portion **816a** and the second flange **812b** includes a second axial flange portion **816b**. The first axial flange portion **816a** extends axially downstream from the first radial flange portion **814a**. The second axial flange portion **816b** extends axially upstream from the second radial flange portion **814b**. In this way, a gap or a space **818** is defined between the first flange **812a** and the second flange **812b**. The space **818** extends axially between the first flange **812a** and the second flange **812b**, as well as radially between the cold side of the hot side component **804** and the axial flange portion **816** of each flange **812**.

[0089] The plurality of flexible plates **850** extend between a fastening mechanism coupling portion **852** and a flange coupling portion **854**. The fastening mechanism coupling portion **852** includes a

fastening mechanism aperture **862** (FIG. **8B**) such that a respective fastening mechanism **820** is inserted through the fastening mechanism aperture **862** (FIG. **8B**). In this way, a respective flexible plate **850** is coupled to a respective fastening mechanism **820**. For example, a first flexible plate **850a** is coupled to the first fastening mechanism **820a** and a second flexible plate **850b** is coupled to the second fastening mechanism **820b**. The first flexible plate **850a** extends from the first fastening mechanism **820a** to the second flange **812b** of the panel **806**. In this way, the first flexible plate **850a** extends in both a radial direction and an axial direction. For example, the first flexible plate **850a** extends radially inwardly at an angle from the first fastening mechanism **820a** towards the second flange **812b** and extends axially downstream from the first fastening mechanism **820a** to the second flange **812b**. The first flexible plate **850a** radially overlaps the second flexible plates **850b** at a substantially axial center of the first flexible plate **850a** and of the second flexible plate **850b**. In this way, the first flexible plate **850a** is disposed in the space **822** between the cold side component **802** and the hot side component **804** and extends through the space **818**.

[0090] Similarly, the second flexible plate **850b** extends from the second fastening mechanism **820b** to the first flange **812a** of the panel **806**. In this way, the second flexible plate **850b** extends in both a radial direction and an axial direction. For example, the second flexible plate **850b** extends radially inward at an angle from the second fastening mechanism **820b** towards the first flange **812a** and extends axially upstream from the second fastening mechanism **820b** to the first flange **812a**. In this way, the second flexible plate **850b** is disposed in the space **822** between the cold side component **802** and the hot side component **804**, and extends through the space **818**.

[0091] Each flexible plate **850** is coupled to a respective flange **812** by the flange coupling portion **854**. For example, the flange coupling portion **854** includes one or more first projections **856** and one or more second projections **858** that are separated at the flange coupling portion **854**, as detailed further below. The one or more second projections **858** are separated from the one or more first projections **856** by one or more slots **857** (FIG. **8B**). The one or more first projections **856** are bent from the flange coupling portion **854** and engage a cold side of the axial flange portion **816**. The one or more second projections **858** are bent to form the flange coupling portion **854** and engage a hot side of the axial flange portion **816**. In this way, the one or more first projections **856** and the one or more second projections **858** together hold or couple the respective axial flange portion **816**. For example, the axial flange portion **816** is press fit between the one or more first projections **856** and the one or more second projections **858**. Accordingly, the first flexible plate **850a** and the second flexible plate **850b** together couple and secure the hot side component **804** (e.g., the panels **806**) to the cold side component **802**.

[0092] The plurality of flexible plates **850** each includes one or more flexible plate cooling holes **860** extending therethrough. The one or more flexible plate cooling holes **860** extend from a cold side to a hot side of the plurality of flexible plates **850** to provide fluid communication through the plurality of flexible plates **850**. The one or more flexible plate cooling holes **860** are disposed in the space **822** when coupling assembly **800** is assembled. In this way, the one or more flexible plate cooling holes **860** provide fluid communication from the space **822** to the one or more hot side component cooling holes **810**. The one or more flexible plate cooling holes **860** can be angled radially and/or circumferentially through the plurality of flexible plates **850** to operably direct cooling air tangentially through the plurality of flexible plates **850**.

[0093] The plurality of flexible plates **850** provide flexibility such that the panels **806** of the hot side component **804** can move with respect to the cold side component **802**. For example, the plurality of flexible plates **850** can compress or can flex such that the panels **806** can move with respect to the cold side component **802**. The plurality of flexible plates **850** are flexible in multiple degrees of freedom such that the panels **806** can move radially, axially, circumferentially, or a combination thereof, with respect to the cold side component **802**. For example, the plurality of flexible plates **850** being circumferentially spaced provides for additional circumferential flexibility compared to coupling assemblies without the benefit of the present disclosure.

[0094] In operation, cooling air **880** is operably directed from the cold side of the cold side component **802**, through the one or more cold side component cooling holes **808**, and into the space **822** between the hot side of the cold side component **802** and the plurality of flexible plates **850**. In the exemplary embodiment, the cooling air **880** is the compressed air **65** (FIG. 2) received from the compressor section of the turbine engine. The cooling air **880**, however, may be supplied from any source, as desired. The cooling air **880** is operably directed from the space **822**, through the one or more flexible plate cooling holes **860**, and between the plurality of flexible plates **850** and the cold side of the hot side component **804**. The cooling air **880** is then operably directed from the space **822** through the one or more hot side component cooling holes **810** and into the combustion chamber **210** to film cool the hot side of the hot side component **804**.

[0095] FIG. 8B is a top view of the plurality of flexible plates **850** isolated from the coupling assembly **800** (FIG. 8A), according to the present disclosure. FIG. 8B shows the plurality of flexible plates **850** in an un-bent orientation such that plurality of flexible plates are flattened. The plurality of flexible plates **850** can be bent or otherwise can be distorted to form the shape shown in FIG. 8A. In the un-bent configuration, the plurality of flexible plates **850** each defines a generally rectangular shape. The plurality of flexible plates **850** may include any shape and/or any size, as desired. The plurality of flexible plates **850** are circumferentially spaced from each other. For example, the first flexible plate **850a** is circumferentially spaced from the second flexible plate **850b** and is positioned circumferentially adjacent the second flexible plate **850b**. In one embodiment, the first flexible plate **850a** is positioned circumferentially aft of the second flexible plate **850b**. The first flexible plate **850a** and the second flexible plate **850b** are generally axially aligned. In this way, the plurality of flexible plates **850** provide balance of the hot side component in the circumferential direction, as well as in the axial direction and in the radial direction. In some embodiments, the first flexible plate **850a** is positioned circumferentially forward of the second flexible plate **850b**.

[0096] The plurality of flexible plates **850** extend from a proximal, first end **851** to a distal, second end **853** opposite the first end **851**. The fastening mechanism coupling portion **852** is located at the first end **851** and the flange coupling portion **854** is located at the second end **853**. The fastening mechanism aperture **862** is located at the fastening mechanism coupling portion **852**. The one or more first projections **856** and the one or more second projections **858** are located at the flange coupling portion **854**. The one or more flexible plate cooling holes **860** are located axially between the fastening mechanism coupling portion **852** and the flange coupling portion **854**.

[0097] Each flexible plate **850** (e.g., the first flexible plate **850a** and the second flexible plate **850b**) includes one or more first bends **870**, one or more second bends **872**, one or more third bends **874**, and one or more fourth bends **876**. FIG. 8B depicts each of the bends by dashed lines. The one or more first bends **870** are defined at the fastening mechanism coupling portion **852** such that the flexible plate **850** extends radially inward from the fastening mechanism coupling portion **852** (shown in FIG. 8A). The one or more second bends **872** are defined at the flange coupling portion **854**. In this way, the flexible plate **850** defines a generally straight portion (e.g., free of bends) between the one or more first bends **870** and the one or more second bends **872**.

[0098] The one or more second bends **872** can include a bend of the one or more first projections **856** in a first radial direction and can include a bend of the one or more second projections **858** in a second radial direction. For example, the one or more first projections **856** can be bent radially outwardly at the one or more second bends **872** and the one or more second projections **858** can be bent radially inwardly at the one or more second bends **872**. In this way, the one or more first projections **856** and the one or more second projections **858** split into different directions at the one or more second bends **872** (as depicted in FIG. 8A). The one or more third bends **874** are defined at the one or more first projections **856** and at the one or more second projections **858**. For example, the one or more first projections **856** are bent radially inwardly at the one or more third bends **874** and the one or more second projections **858** are bent radially outwardly at the one or more third

bends **874** (as depicted in FIG. **8A**). In this way, the one or more first projections **856** include a portion that engages with the cold side of the axial flange portion **816**, and the one or more second projections **858** include a portion that engages with the hot side of the axial flange portion **816**. The one or more fourth bends **876** are defined at the one or more first projections **856** and at the one or more second projections **858**. For example, the one or more first projections **856** are bent radially outwardly at the one or more fourth bends **876** and the one or more second projections **858** are bent radially inwardly at the one or more fourth bends **876**. In this way, the one or more first projections **856** and the one or more second projections **858** engage the axial flange portion **816** at the one or more fourth bends **876** (as depicted in FIG. **8A**). Thus, the bends of the flange coupling portion **854** allow for the one or more first projections **856** and the one or more second projections **858** to engage and to hold the panel **806** at the axial flange portion **816**. The plurality of flexible plates **850** may include any number of bends, in any orientation, as desired, for engaging the axial flange portion **816** of the panel **806**.

[0099] Referring again to FIG. **8A**, the coupling assembly **800** provides for a cooling arrangement that avoids direct exposure of the plurality of fastening mechanisms **820** to the hot combustion gases in the combustion chamber **210**. The coupling assembly **800** provides for separating the plurality of fastening mechanisms **820** from the hot side component **804** such that the plurality of fastening mechanisms **820** are not exposed to the hot combustion gases. For example, the plurality of fastening mechanisms **820** are insulated from the combustion chamber **210**. The plurality of flexible plates **850** engage the panel **806** of the hot side component **804** such that the hot side component **804** is coupled to the cold side component **802** without the need of the fastening mechanisms **820** to directly contact the hot side component **804**. Therefore, the thermal gradients in an area of plurality of fastening mechanisms **820** are reduced, thereby reducing the stress concentration on the plurality of fastening mechanisms **820**. The flexibility of the plurality of flexible plates **850** allows the panels **806** to move while also reducing the stress on the plurality of fastening mechanisms **820** compared to arrangements in which the fastening mechanisms **820** are directly contacting the hot side component **804**. The projections of flange coupling portion **854** of the flexible plates **850** also includes flexibility such that the flange coupling portion **854** provides frictional damping when the hot side component **804** moves with respect to the cold side component **802**. Thus, the coupling assembly **800** provides for reducing damage and/or for reducing failure of the hot side component **804** and/or of the cold side component **802** in an area about the plurality of fastening mechanisms **820**.

[0100] FIG. **9A** is a schematic partial cross-sectional view of a portion of a coupling assembly **900** for a combustion section **26** (FIGS. **1** and **2**) of a turbine engine **10** (FIG. **1**), taken at a centerline axis of the turbine engine **10** (FIG. **1**), according to another embodiment. The coupling assembly **900** is substantially similar to the coupling assembly **800** of FIG. **8A**. For example, the coupling assembly **900** includes the cold side component **802**, the hot side component **804**, the panels **806**, the plurality of fastening mechanisms **820**, and the plurality of flanges **812**. The coupling assembly **900**, however, includes a plurality of flexible plates **950** that are different than the plurality of flexible plates **850** of FIGS. **8A** and **8B**. FIG. **9B** is a top view of the plurality of flexible plates **950** isolated from the coupling assembly **900** (FIG. **9A**), according to the present disclosure.

[0101] With reference to FIGS. **9A** and **9B**, the plurality of flexible plates **950** include a first flexible plate **950a** and a second flexible plate **950b**. The plurality of flexible plates **950** extend between a fastening mechanism coupling portion **952** and a flange coupling portion **954**. The plurality of flexible plates **950** include a plurality of fastening mechanism apertures **962**. For example, the fastening mechanism coupling portion **952** includes a first fastening mechanism aperture **962a** such that a respective fastening mechanism **820** is inserted through the first fastening mechanism aperture **962a**. In this way, a respective flexible plate **950** is coupled to a respective fastening mechanism **820**. For example, the first flexible plate **950a** is coupled to the first fastening mechanism **820a** and the second flexible plate **950b** is coupled to the second fastening mechanism

820b. The flange coupling portion **954** includes a second fastening mechanism aperture **962b**. In this way, the plurality of flexible plates **950** can be oriented such that the respective fastening mechanisms **820** are inserted through the first fastening mechanism aperture **962a** or can be oriented such that the respective fastening mechanisms **820** are inserted through the second fastening mechanism aperture **962b**. Such an arrangement provides for ease of manufacturing while the coupling assembly **900** is being assembled.

[0102] The first flexible plate **950a** extends from the first fastening mechanism **820a** to the second flange **812b** of the panel **806**, as detailed above. The first flexible plate **950a** radially overlaps the second flexible plates **950b** at a substantially axial center of the first flexible plate **950a** and of the second flexible plate **950b** (shown in FIG. 9A). In this way, the first flexible plate **950a** is disposed in the space **822** between the cold side component **802** and the hot side component **804** and extends through the space **818**. Similarly, the second flexible plate **950b** extends from the second fastening mechanism **820b** to the first flange **812a** of the panel **806**, as detailed above. In this way, the second flexible plate **950b** is disposed in the space **822** between the cold side component **802** and the hot side component **804**, and extends through the space **818**.

[0103] Each flexible plate **950** is coupled to a respective flange **812** by the flange coupling portion **954**. The flange coupling portion **954**, however, does not include one or more projections like the flange coupling portion **854** detailed with respect to FIG. 8A. Rather, the flange coupling portion **954** is inserted in the space **818** to engage with the hot side of the axial flange portion **816** such that flange coupling portion **954** radially supports the panel **806** at the axial flange portion **816**. The flange coupling portion **954** is secured to the axial flange portion **816** by, for example, welding, brazing, adhesive, or the like. Accordingly, the first flexible plate **950a** and the second flexible plate **950b** together couple and secure the hot side component **804** (e.g., the panels **806**) to the cold side component **802** such that the plurality of fastening mechanisms **820** do not directly engage the hot side component **804**. In this way, the fastening mechanisms **820** are insulated from the combustion chamber **210**.

[0104] The plurality of flexible plates **950** each includes one or more flexible plate cooling holes **960** extending therethrough. The one or more flexible plate cooling holes **960** extend from a cold side to a hot side of the plurality of flexible plates **950** to provide fluid communication through the plurality of flexible plates **950**. The one or more flexible plate cooling holes **960** are disposed in the space **822** when coupling assembly **900** is assembled. In this way, the one or more flexible plate cooling holes **960** provide fluid communication from the space **822** to the one or more hot side component cooling holes **810**. The one or more flexible plate cooling holes **960** can be angled radially and/or circumferentially through the plurality of flexible plates **950** to operably direct cooling air tangentially through the plurality of flexible plates **850**.

[0105] FIG. 9B shows the first flexible plate **950a** is circumferentially spaced from the second flexible plate **850b** and is positioned circumferentially adjacent the second flexible plate **950b**. For example, the first flexible plate **950a** is positioned circumferentially aft of the second flexible plate **950b**. In this way, the plurality of flexible plates **950** are circumferentially spaced from each other. The first flexible plate **950a** and the second flexible plate **950b** are generally axially aligned. In this way, the plurality of flexible plates **950** provide balance of hot side component **804** (FIG. 9A) in the circumferential direction, as well as in the axial direction and in the radial direction. In some embodiments, the first flexible plate **950a** is positioned circumferentially forward of the second flexible plate **950b**.

[0106] The plurality of flexible plates **950** extend from a proximal, first end **951** (FIG. 9B) to a distal, second end **953** (FIG. 9B) opposite the first end **951**. The fastening mechanism coupling portion **952** is located at the first end **951** and the flange coupling portion **954** is located at the second end **953**. The first fastening mechanism aperture **962a** is located at the fastening mechanism coupling portion **952** and the second fastening mechanism aperture **962b** is located at the flange coupling portion **954**. FIG. 9B shows the one or more flexible plate cooling holes **960** are located

axially between the fastening mechanism coupling portion **952** and the flange coupling portion **954**.

[0107] Each flexible plate **950** (e.g., the first flexible plate **950a** and the second flexible plate **950b**) includes a first bend **970** and a second bend **972**. FIG. **9B** depicts each of the bends by dashed lines. The first bend **970** is defined at the fastening mechanism coupling portion **952** such that the flexible plate **950** extends radially inward from the fastening mechanism coupling portion **952** (shown in FIG. **9A**). The second bend **972** is defined at the flange coupling portion **954** such that the flexible plate **950** extends from the second bend **972** generally axially. In this way, the flexible plate **950** defines a generally straight portion (e.g., free of bends) between the first bend **970** and the second bend **972**. The portion of the flexible plate **950** that extends axially from the second bend **972** is inserted into the space **818** (FIG. **9A**) and engages the cold side of the axial flange portion **816** to support the panel **806** of the hot side component **804**.

[0108] The plurality of flexible plates **950** provide flexibility such that the panels **806** of the hot side component **804** can move with respect to the cold side component **802**. For example, the plurality of flexible plates **950** can compress or can flex such that the panels **806** can move with respect to the cold side component **802**. The plurality of flexible plates **950** are flexible in multiple degrees of freedom such that the panels **806** can move radially, axially, circumferentially, or a combination thereof, with respect to the cold side component **802**.

[0109] In operation, cooling air **880** is operably directed from the cold side of the cold side component **802**, through the one or more cold side component cooling holes **808**, and into the space **822** between the hot side of the cold side component **802** and the plurality of flexible plates **950**. The cooling air **880** is operably directed from the space **822**, through the one or more flexible plate cooling holes **960**, and between the plurality of flexible plates **950** and the cold side of the hot side component **804**. The cooling air **880** is then operably directed from the space **822** through the one or more hot side component cooling holes **810** and into the combustion chamber **210** to film cool the hot side of the hot side component **804**. The plurality of flexible plates **950** provides for ease of manufacturing compared to the plurality of flexible plates **850** of FIGS. **8A** and **8B**. The coupling of the plurality of flexible plates **950** to the axial flange portion **816** reduces wear due to the flange coupling portion **954** being coupled to the axial flange portion **816** such that the flange coupling portion **954** does not slide or move with respect to the axial flange portion **816**.

[0110] The embodiments of the present disclosure provide for improved cooling arrangements such that the one or more fastening mechanisms of the combustion section are not directly exposed to the hot combustion gases in the combustion chamber. Thus, the embodiments of the present disclosure help to reduce the thermal gradients in an area about the fastening mechanisms, thereby reducing a stress concentration in the area about the fastening mechanisms. Accordingly, the embodiments of the present disclosure provide for reducing damage and/or for reducing failure of the hot side component and/or of the cold side component in an area about the fastening mechanisms.

[0111] Further aspects are provided by the subject matter of the following clauses.

[0112] A coupling assembly for a turbine engine comprises a cold side component, a hot side component, the cold side component and the hot side component together at least partially forming a combustion chamber, and a fastening mechanism coupling the hot side component to the cold side component, the fastening mechanism comprising a stud disposed through the cold side component, and a cap positioned on the stud, the cap defining a hollow interior and including one or more first cap cooling holes that operably direct cooling air into the hollow interior such that the hollow interior provides a cushion of cooling air between the combustion chamber and the stud.

[0113] The coupling assembly of the preceding clause, a space being defined between the cold side component and the hot side component, the stud being disposed in the space.

[0114] The coupling assembly of any preceding clause, the cap being disposed through

[0115] the hot side component.

[0116] The coupling assembly of any preceding clause, the cap being positioned between the stud and the combustion chamber.

[0117] The coupling assembly of any preceding clause, the stud being radially spaced from the combustion chamber such that the stud is insulated from the combustion chamber.

[0118] The coupling assembly of any preceding clause, cap including one or more second cap cooling holes that operably direct the cooling air from the hollow interior into the combustion chamber.

[0119] The coupling assembly of any preceding clause, the hollow interior including a diameter that is larger than an outer diameter of the stud.

[0120] The coupling assembly of any preceding clause, the cap including a thin wall defined by an interior surface and an exterior surface, the thin wall defining a shape of the hollow interior.

[0121] The coupling assembly of any preceding clause, the shape of the hollow interior generally corresponding to a shape of the cap.

[0122] The coupling assembly of any preceding clause, further comprising a washer disposed between the cold side component and the hot side component, the washer being flexible such that the hot side component moves with respect to the cold side component and the washer prevents the hot side component from contacting the cold side component.

[0123] The coupling assembly of any preceding clause, the washer including one or more washer cooling holes that operably direct cooling air towards the cap.

[0124] The coupling assembly of any preceding clause, the washer including one or more washer slots that extend radially inward from an outer edge of the washer to the one or more washer cooling holes.

[0125] The coupling assembly of any preceding clause, the one or more first cap cooling holes operably directing the cooling air from the space to the hollow interior.

[0126] The coupling assembly of any preceding clause, the one or more cap including an exterior surface and an interior surface, the one or more first cap cooling holes extending through the thin wall from the exterior surface to the interior surface.

[0127] The coupling assembly of any preceding clause, the cold side component being a shell of a liner of the turbine engine, and the hot side component being a heat shield.

[0128] The coupling assembly any preceding clause, the cold side component being an annular dome assembly, and the hot side component being a deflector assembly.

[0129] The coupling assembly of any preceding clause, the cold side component including one or more cold side component cooling holes that operably direct the cooling air from a cold side of the cold side component to the space between the cold side component and the hot side component.

[0130] The coupling assembly of any preceding clause, the one or more cold side component cooling holes being impingement cooling holes.

[0131] The coupling assembly of any preceding clause, the hot side component including one or more hot side component cooling holes that operably direct the cooling air from the space to the combustion chamber.

[0132] The coupling assembly of any preceding clause, the one or more hot side component cooling holes being film cooling holes.

[0133] The coupling assembly of any preceding clause, the stud being inserted through a first fastening mechanism aperture of the cold side component, and the cap being inserted through a second fastening mechanism aperture of the hot side component.

[0134] The coupling assembly of any preceding clause, the hot side component including one or more panels that together form the hot side component.

[0135] The coupling assembly of any preceding clause, each panel of the one or more panels being coupled to the cold side component by one or more fastening mechanisms.

[0136] The coupling assembly of any preceding clause, the cap being coupled to the stud by a first threaded connection.

[0137] The coupling assembly of any preceding clause, the cap being press fit onto the stud.

[0138] The coupling assembly of any preceding clause, the cap being formed integral with the stud.

[0139] The coupling assembly of any preceding clause, the fastening assembly further comprising a nut that is coupled to the stud on a cold side of the stud.

[0140] The coupling assembly of any preceding clause, the nut being coupled to the stud by a second threaded connection.

[0141] The coupling assembly of any preceding clause, the cap comprising a stud receiving portion that defines a stud opening that receives a portion of the stud.

[0142] The coupling assembly of any preceding clause, the stud receiving portion extending from a cold side of an interior surface of the cap and into the hollow interior.

[0143] The coupling assembly of any preceding clause, the stud receiving portion being spaced from a hot side of the interior surface of the cap.

[0144] The coupling assembly of any preceding clause, the cap further comprising a cap insert positioned in the stud opening.

[0145] The coupling assembly of any preceding clause, the stud being coupled to the cap by the cap insert.

[0146] The coupling assembly of any preceding clause, the stud being coupled to the cap insert by the first threaded connection.

[0147] The coupling assembly of any preceding clause, the cap being press fit onto the cap insert.

[0148] The coupling assembly of any preceding clause, the cap insert being formed integrally with the cap.

[0149] The coupling assembly of any preceding clause, the stud being press fit onto the cap insert.

[0150] The coupling assembly of any preceding clause, the stud being formed integrally with the cap insert.

[0151] The coupling assembly of any preceding clause, the stud being brazed or welded to the cap insert.

[0152] The coupling assembly of any preceding clause, the washer including a flat portion that contacts the cold side component and a bend portion that contacts the hot side component.

[0153] The coupling assembly of any preceding clause, the washer including a washer stud hole for receiving the stud.

[0154] The coupling assembly of any preceding clause, the one or more washer cooling holes being disposed in the bend portion of the washer.

[0155] The coupling assembly of any preceding clause, the one or more washer cooling holes being disposed within the space between the cold side component and the hot side component.

[0156] The coupling assembly of any preceding clause, the cap defining a radial space about the stud receiving portion.

[0157] The coupling assembly of any preceding clause, the radial space extending radially from an exterior surface of the stud receiving portion to another portion of the exterior surface of the cap.

[0158] The coupling assembly of any preceding clause, the exterior surface defining a generally U-shaped portion that defines the radial space.

[0159] The coupling assembly of any preceding clause, the stud receiving portion abutting the washer.

[0160] The coupling assembly of any preceding clause, the cap being radially spaced from the washer such that the cooling air is operably directed between the washer and the cap and into the radial space.

[0161] The coupling assembly of any preceding clause, the cap further including one or more third cap cooling holes that operably direct the cooling air from the radial space into the hollow interior.

[0162] The coupling assembly of any preceding clause, the hot side component including a recessed portion in an area about the fastening mechanism.

[0163] The coupling assembly of any preceding clause, the recessed portion being recessed with

respect to the hot side of the hot side component.

[0164] The coupling assembly of any preceding clause, the recessed portion extending from the hot side component towards the cold side component.

[0165] The coupling assembly of any preceding clause, the recessed portion defining a recessed portion space, the cap being disposed in the recessed portion space.

[0166] The coupling assembly of any preceding clause, the washer having a diameter greater than a diameter of the recessed portion such that the washer circumferentially surrounds the recessed portion.

[0167] The coupling assembly of any preceding clause, the recessed portion including one or more recessed portion cooling holes that operably direct the cooling air from the space to the combustion chamber.

[0168] The coupling assembly of any preceding clause, a portion of the cap being spaced from the cold side of the recessed portion such that a space is defined between the cap and the recessed portion.

[0169] The coupling assembly of any preceding clause, the cap further including a cap extension that is inserted into a cap receiving portion of the stud.

[0170] The coupling assembly of any preceding clause, the recessed portion comprising a stud receiving portion for receiving the stud.

[0171] The coupling assembly of any preceding clause, the stud including one or more stud cooling holes that operably direct the cooling from the cold side of the cold side component to the hollow interior of the cap.

[0172] The coupling assembly of any preceding clause, the cold side of the cap abutting the hot side of the recessed portion.

[0173] The coupling assembly of any preceding clause, the cap being coupled to the stud

[0174] by the cap extension.

[0175] The coupling assembly of any preceding clause, the hot side of the cap being substantially flush with the hot side of the hot side component.

[0176] The coupling assembly of any preceding clause, the cap comprising one or more third cap cooling holes that operably direct the cooling air from the one or more stud cooling holes to the hollow interior.

[0177] The coupling assembly of any preceding clause, a cold side of the cap being coupled to the stud.

[0178] The coupling assembly of any preceding clause, the stud comprising a stud head, the cap being coupled to the stud head.

[0179] The coupling assembly of any preceding clause, the cap being coupled to the stud by welding, brazing, or adhesive.

[0180] The coupling assembly of any preceding clause, the cap being formed integrally with the stud.

[0181] A turbine engine comprises a combustion section including a combustion chamber and a coupling assembly comprising a cold side component, a hot side component, the cold side component and the hot side component together at least partially forming the combustion chamber, and a fastening mechanism coupling the hot side component to the cold side component, the fastening mechanism comprising a stud disposed through the cold side component, and a cap positioned on the stud, the cap defining a hollow interior and including one or more first cap cooling holes that operably direct cooling air into the hollow interior such that the hollow interior provides a cushion of cooling air between the combustion chamber and the stud.

[0182] The turbine engine of the preceding clause, the cap being disposed through the hot side component.

[0183] The turbine engine of any preceding clause, the cap being positioned between the stud and the combustion chamber.

[0184] The turbine engine of any preceding clause, the stud being spaced radially from the combustion chamber such that the stud is insulated from the combustion chamber.

[0185] The turbine engine of any preceding clause, cap including one or more second cap cooling holes that operably direct the cooling air from the hollow interior into the combustion chamber.

[0186] The turbine engine of any preceding clause, the hollow interior including a diameter that is larger than an outer diameter of the stud.

[0187] The turbine engine of any preceding clause, the cap including a thin wall defined by an interior surface and an exterior surface, the thin wall defining a shape of the hollow interior.

[0188] The turbine engine of any preceding clause, the shape of the hollow interior generally corresponding to a shape of the cap.

[0189] The turbine engine of any preceding clause, a space being defined between the cold side component and the hot side component, the stud being disposed in the space.

[0190] The turbine engine of any preceding clause, further comprising a washer disposed between the cold side component and the hot side component, the washer being flexible such that the hot side component moves with respect to the cold side component and the washer prevents the hot side component from contacting the cold side component.

[0191] The turbine engine of any preceding clause, the washer including one or more washer cooling holes that operably direct cooling air towards the cap.

[0192] The turbine engine of any preceding clause, the washer including one or more washer slots that extend radially inward from an outer edge of the washer to the one or more cooling holes.

[0193] The turbine engine of any preceding clause, the one or more first cap cooling holes operably directing the cooling air from the space to the hollow interior.

[0194] The turbine engine of any preceding clause, the one or more cap including an exterior surface and an interior surface, the one or more first cap cooling holes extending through the thin wall from the exterior surface to the interior surface.

[0195] The turbine engine of any preceding clause, the cold side component being a shell of a liner of the turbine engine, and the hot side component being a heat shield.

[0196] The turbine engine of any preceding clause, the cold side component being an annular dome assembly, and the hot side component being a deflector assembly.

[0197] The turbine engine of any preceding clause, the cold side component including one or more cold side component cooling holes that operably direct the cooling air from a cold side of the cold side component to the space between the cold side component and the hot side component.

[0198] The turbine engine of any preceding clause, the one or more cold side component cooling holes being impingement cooling holes.

[0199] The turbine engine of any preceding clause, the hot side component including one or more hot side component cooling holes that operably direct the cooling air from the space to the combustion chamber.

[0200] The turbine engine of any preceding clause, the one or more hot side component cooling holes being film cooling holes.

[0201] The turbine engine of any preceding clause, the stud being inserted through a first fastening mechanism aperture of the cold side component, and the cap being inserted through a second fastening mechanism aperture of the hot side component.

[0202] The turbine engine of any preceding clause, the hot side component including one or more panels that together form the hot side component.

[0203] The turbine engine of any preceding clause, each panel of the one or more panels being coupled to the cold side component by one or more fastening mechanisms.

[0204] The turbine engine of any preceding clause, the cap being coupled to the stud by a first threaded connection.

[0205] The turbine engine of any preceding clause, the cap being press fit onto the stud.

[0206] The turbine engine of any preceding clause, the cap being formed integrally with the stud.

[0207] The turbine engine of any preceding clause, the fastening assembly further comprising a nut that is coupled to the stud on a cold side of the stud.

[0208] The turbine engine of any preceding clause, the nut being coupled to the stud by a second threaded connection.

[0209] The turbine engine of any preceding clause, the cap comprising a stud receiving portion that defines a stud opening that receives a portion of the stud.

[0210] The turbine engine of any preceding clause, the stud receiving portion extending from a cold side of an interior surface of the cap and into the hollow interior.

[0211] The turbine engine of any preceding clause, the stud receiving portion being spaced from a hot side of the interior surface of the cap.

[0212] The turbine engine of any preceding clause, the cap further comprising a cap insert positioned in the stud opening.

[0213] The turbine engine of any preceding clause, the stud being coupled to the cap by the cap insert.

[0214] The turbine engine of any preceding clause, the stud being coupled to the cap insert by the first threaded connection.

[0215] The turbine engine of any preceding clause, the cap being press fit onto the cap insert.

[0216] The turbine engine of any preceding clause, the cap insert being formed integrally with the cap.

[0217] The turbine engine of any preceding clause, the stud being press fit onto the cap insert.

[0218] The turbine engine of any preceding clause, the stud being formed integrally with the cap insert.

[0219] The turbine engine of any preceding clause, the stud being brazed or welded to the cap insert.

[0220] The turbine engine of any preceding clause, the washer including a flat portion that contacts the cold side component and a bend portion that contacts the hot side component.

[0221] The turbine engine of any preceding clause, the washer including a washer stud hole for receiving the stud.

[0222] The turbine engine of any preceding clause, the one or more washer cooling holes being disposed in the bend portion of the washer.

[0223] The turbine engine of any preceding clause, the one or more washer cooling holes being disposed within the space between the cold side component and the hot side component.

[0224] The turbine engine of any preceding clause, the cap defining a radial space about the stud receiving portion.

[0225] The turbine engine of any preceding clause, the radial space extending radially from an exterior surface of the stud receiving portion to another portion of the exterior surface of the cap.

[0226] The turbine engine of any preceding clause, the exterior surface defining a generally U-shaped portion that defines the radial space.

[0227] The turbine engine of any preceding clause, the stud receiving portion abutting the washer.

[0228] The turbine engine of any preceding clause, the cap being radially spaced from the washer such that the cooling air is operably directed between the washer and the cap and into the radial space.

[0229] The turbine engine of any preceding clause, the cap further including one or more third cap cooling holes that operably direct the cooling air from the radial space into the hollow interior.

[0230] The turbine engine of any preceding clause, the hot side component including a recessed portion in an area about the fastening mechanism.

[0231] The coupling assembly of any preceding clause, the recessed portion being recessed with respect to the hot side of the hot side component.

[0232] The turbine engine of any preceding clause, the recessed portion extending from the hot side component towards the cold side component.

[0233] The turbine engine of any preceding clause, the recessed portion defining a recessed portion space, the cap being disposed in the recessed portion space.

[0234] The turbine engine of any preceding clause, the washer having a diameter greater than a diameter of the recessed portion such that the washer circumferentially surrounds the recessed portion.

[0235] The turbine engine of any preceding clause, the recessed portion including one or more recessed portion cooling holes that operably direct the cooling air from the space to the combustion chamber.

[0236] The turbine engine of any preceding clause, a portion of the cap being spaced from the cold side of the recessed portion such that a space is defined between the cap and the recessed portion.

[0237] The turbine engine of any preceding clause, the cap further including a cap extension that is inserted into a cap receiving portion of the stud.

[0238] The turbine engine of any preceding clause, the recessed portion comprising a stud receiving portion for receiving the stud.

[0239] The turbine engine of any preceding clause, the stud including one or more stud cooling holes that operably direct the cooling from the cold side of the cold side component to the hollow interior of the cap.

[0240] The turbine engine of any preceding clause, the cold side of the cap abutting the hot side of the recessed portion.

[0241] The turbine engine of any preceding clause, the cap being coupled to the stud by the cap extension.

[0242] The turbine engine of any preceding clause, the hot side of the cap being substantially flush with the hot side of the hot side component.

[0243] The turbine engine of any preceding clause, the cap comprising one or more third cap cooling holes that operably direct the cooling air from the one or more stud cooling holes to the hollow interior.

[0244] The turbine engine of any preceding clause, a cold side of the cap being coupled to the stud.

[0245] The turbine engine of any preceding clause, the stud comprising a stud head, the cap being coupled to the stud head.

[0246] The turbine engine of any preceding clause, the cap being coupled to the stud by welding, brazing, or adhesive.

[0247] The turbine engine of any preceding clause, the cap being formed integrally with the stud.

[0248] A method of operably causing cooling air to flow through a turbine engine. The method comprises operably flowing cooling air through a cold side component of the turbine engine, a hot side component being coupled to the cold side component by a fastening mechanism that comprises a stud disposed through the cold side component and a cap, the cold side component and the hot side component together at least partially forming a combustion chamber. The method comprises operably directing the cooling air through one or more first cap cooling holes in the cap into a hollow interior of the cap such that the hollow interior provides a cushion of cooling air between the combustion chamber and the stud.

[0249] The method of the preceding clause, further comprising operably directing the cooling air through the hot side component.

[0250] The method of any preceding clause, further comprising operably directing the cooling air through one or more second cap cooling holes from the hollow interior to the combustion chamber.

[0251] The method of any preceding clause, further comprising operably directing the cooling air through one or more washer cooling holes of a washer disposed between the cold side component and the hot side component, the washer being flexible such that the hot side component moves with respect to the cold side component and the washer prevents the hot side component from contacting the cold side component.

[0252] The method of any preceding clause, further comprising operably directing the cooling air

through the one or more washer cooling holes towards the cap.

[0253] The method of any preceding clause, further comprising operably directing the cooling air through one or more cold side component cooling holes in the cold side component and into a space defined between the cold side component and the hot side component.

[0254] The method of any preceding clause, further comprising operably directing the cooling air from the space to the hollow interior.

[0255] The method of any preceding clause, further comprising operably directing the cooling air through one or more hot side component cooling holes in the hot side component and into the combustion chamber.

[0256] The method of any preceding clause, further comprising operably directing the cooling air into a radial space between the washer and the cap.

[0257] The method of any preceding clause, further comprising operably directing the cooling air from the radial space into the hollow interior through one or more third cap cooling holes.

[0258] The method of any preceding clause, the hot side component including a recessed portion, the method further comprising operably directing the cooling air through one or more recessed portion cooling holes in the recessed portion from the space to the combustion chamber.

[0259] The method of any preceding clause, the cap being disposed in a recessed portion space defined by the recessed portion, the method further comprising operably directing the cooling air from the recessed portion space through the one or more first cap cooling holes and into the hollow interior of the cap.

[0260] The method of any preceding clause, the stud comprising one or more stud cooling holes, the method further comprising operably directing the cooling air from the cold side of the cold side component through the one or more stud cooling holes.

[0261] The method of any preceding clause, the cap further comprising one or more third cap cooling holes, the method further comprising operably directing the cooling air from the one or more stud cooling holes through the one or more third cap cooling holes and into the hollow interior.

[0262] The method of any preceding clause, further comprising operably directing the cooling air through the washer and into a space between the washer and the cap.

[0263] The method of any preceding clause, further comprising operably directing the cooling air through the washer and into a space between the washer and the recessed portion.

[0264] The method of any preceding clause, further comprising film cooling the cap by operably directing the cooling air from the hollow interior through the one or more second cap cooling holes and into the combustion chamber.

[0265] The method of any preceding clause, further comprising impingement cooling the hot side component by operably directing the cooling air from the cold side of the cold side component through the one or more cold side component cooling holes and into the space between the cold side component and the hot side component.

[0266] The method of any preceding clause, the cooling air being compressed air from a compressor section of the turbine engine.

[0267] The method of any preceding clause, the cold side component being a shell of a liner of the turbine engine, and the hot side component being a heat shield.

[0268] The method of any preceding clause, the cold side component being an annular dome assembly, and the hot side component being a deflector assembly.

[0269] A coupling assembly for a turbine engine comprises a cold side component, a hot side component, the cold side component and the hot side component together at least partially forming a combustion chamber, a plurality of fastening mechanisms disposed through the cold side component, and a plurality of flexible plates, each of the plurality of flexible plates being circumferentially spaced from each other and extending from a respective one of the plurality of fastening mechanisms to the hot side component and engaging the hot side component to couple

the hot side component to the cold side component such that the plurality of fastening mechanisms are insulated from the combustion chamber.

[0270] The coupling assembly of the preceding clause, each of the plurality of flexible plates including one or more flexible plate cooling holes that operably direct cooling air through the plurality of flexible plates.

[0271] The coupling assembly of any preceding clause, the plurality of fastening mechanisms comprising a first fastening mechanism and a second fastening mechanism, and the plurality of flexible plates comprise a first flexible plate extending from the first fastening mechanism to the hot side component and a second flexible plate extending from the second fastening mechanism to the hot side component.

[0272] The coupling assembly of any preceding clause, the first fastening mechanism being located upstream of the second fastening mechanism.

[0273] The coupling assembly of any preceding clause, the first flexible plate extending from the first fastening mechanism downstream and to the hot side component.

[0274] The coupling assembly of any preceding clause, the second flexible plate extending from the second fastening mechanism upstream and to the hot side component.

[0275] The coupling assembly of any preceding clause, the hot side component comprising a plurality of flanges, each of the plurality of flexible plates engaging a respective one of the plurality of flanges to couple the hot side component to the cold side component.

[0276] The coupling assembly of any preceding clause, each of the plurality of flanges comprising a radial flange portion extending substantially radially from the hot side component and an axial flange portion extending substantially axially from the radial flange portion.

[0277] The coupling assembly of any preceding clause, each of the plurality of flexible plates engaging a respective axial flange portion.

[0278] The coupling assembly of any preceding clause, the plurality of flanges comprising a first flange and a second flange positioned downstream of the first flange, and the plurality of flexible plates comprise a first flexible plate and a second flexible plate, the first flexible plate engaging the second flange and the second flexible plate engaging the first flange.

[0279] The coupling assembly of any preceding clause, further comprising a space defined between the first flange and the second flange, each of the plurality of flexible plates extending into the space to couple the hot side component to the cold side component.

[0280] The coupling assembly of any preceding clause, the first flange comprising a first radial flange portion and a first axial flange portion, the first axial flange portion extending axially downstream from the first radial flange portion.

[0281] The coupling assembly of any preceding clause, the second flange comprising a second radial flange portion and a second axial flange portion, the second axial flange portion extending axially upstream from the second radial flange portion.

[0282] The coupling assembly of any preceding clause, the hot side component comprising one or more panels, the one or more panels being coupled to the cold side component by the plurality of flexible plates.

[0283] The coupling assembly of any preceding clause, the plurality of flexible plates each comprising a fastening mechanism aperture for receiving a respective fastening mechanism of the one or more fastening mechanisms.

[0284] The coupling assembly of any preceding clause, the plurality of flexible plates each comprising a fastening coupling portion that is coupled to a respective fastening mechanism of the one or more fastening mechanisms and a flange coupling portion that is coupled to a respective flange of the plurality of flanges.

[0285] The coupling assembly of any preceding clause, the flange coupling portion comprising one or more first projections and one or more second projections that are separated at the flange coupling portion, the one or more first projections and the one or more second projections engaging

with a respective flange of the plurality of flanges.

[0286] The coupling assembly of any preceding clause, the one or more first projections engaging a hot side of the respective flange, and the one or more second projections engaging a cold side of the respective flange.

[0287] The coupling assembly of any preceding clause, the respective flange being press fit between the one or more first projections and the one or more second projections.

[0288] The coupling assembly of any preceding clause, the plurality of flexible plates each comprising a first bend defined at the fastening mechanism coupling portion such that the plurality of flexible plates extend radially inward from the fastening mechanism coupling portion.

[0289] The coupling assembly of any preceding clause, the plurality of flexible plates each comprising one or more second bends defined at the flange coupling portion, the one or more first projections and the one or more second projections being bent at the one or more second bends.

[0290] The coupling assembly of any preceding clause, the one or more first projections being bent radially outwardly at the one or more second bends, and the one or more second projections being bent radially inwardly at the one or more second bends.

[0291] The coupling assembly of any preceding clause, the one or more first projections and the one or more second projections being split into different directions at the one or more second bends.

[0292] The coupling assembly of any preceding clause, the plurality of flexible plates each comprising one or more third bends at the one or more first projections and the one or more second projections.

[0293] The coupling assembly of any preceding clause, the one or more first projections being bent radially inwardly at the one or more third bends, and the one or more second projections being bent radially outwardly at the one or more third bends.

[0294] The coupling assembly of any preceding clause, the plurality of flexible plates each comprising one or more fourth bends at the one or more first projections and the one or more second projections.

[0295] The coupling assembly of any preceding clause, the one or more first projections being bent radially outwardly at the one or more fourth bends, and the one or more second projections being bent radially inwardly at the one or more fourth bends.

[0296] The coupling assembly of any preceding clause, the one or more first projections and the one or more second projections engaging the axial flange portion at the one or more fourth bends.

[0297] The coupling assembly of any preceding clause, the plurality of flexible plates each comprising a first fastening mechanism aperture and a second fastening mechanism aperture.

[0298] The coupling assembly of any preceding clause, the first fastening mechanism aperture being located at the fastening mechanism coupling portion, and the second fastening mechanism aperture being located at the flange coupling portion.

[0299] The coupling assembly of any preceding clause, the flange coupling portion engaging a cold side of the axial flange portion.

[0300] The coupling assembly of any preceding clause, the plurality of flexible plates each including a first bend at the fastening mechanism coupling portion and a second bend at the flange coupling portion.

[0301] The coupling assembly of any preceding clause, the flange coupling portion holding the hot side component at the axial flange portion.

[0302] The coupling assembly of any preceding clause, the flange coupling portion being secured to the axial flange portion by at least one of brazing, welding, or adhesive.

[0303] The coupling assembly of any preceding clause, the plurality of flexible plates being flexible in multiple degrees of freedom such that the hot side component moves axially, radially, or circumferentially with respect to the cold side component.

[0304] The coupling assembly of any preceding clause, the first flexible plate being positioned

circumferentially adjacent the second flexible plate.

[0305] The coupling assembly of any preceding clause, the first flexible plate being positioned circumferentially aft with respect to the second flexible plate.

[0306] The coupling assembly of any preceding clause, a space being defined between the cold side component and the hot side component, the plurality of plates being disposed in the space.

[0307] The coupling assembly of any preceding clause, the one or more fastening mechanisms being disposed in the space between the cold side component and the hot side component.

[0308] The coupling assembly of any preceding clause, the one or more fastening mechanisms comprising a stud and a stud head, the stud head being disposed on a hot side of the cold side component.

[0309] The coupling assembly of any preceding clause, the stud head being disposed in the space between the cold side component and the hot side component.

[0310] The coupling assembly of any preceding clause, the cold side component being a shell of a liner of the turbine engine, and the hot side component being a heat shield.

[0311] The coupling assembly any preceding clause, the cold side component being an annular dome assembly, and the hot side component being a deflector assembly.

[0312] The coupling assembly of any preceding clause, the cold side component including one or more cold side component cooling holes that operably direct the cooling air from a cold side of the cold side component to the space between the cold side component and the hot side component.

[0313] The coupling assembly of any preceding clause, the one or more cold side component cooling holes being impingement cooling holes.

[0314] The coupling assembly of any preceding clause, the hot side component including one or more hot side component cooling holes that operably direct the cooling air from the space to the combustion chamber.

[0315] The coupling assembly of any preceding clause, the one or more hot side component cooling holes being film cooling holes.

[0316] The coupling assembly of any preceding clause, the one or more fastening mechanisms each comprising a stud extending through the cold side component.

[0317] The coupling assembly of any preceding clause, the fastening assembly further comprising a nut that is coupled to the stud on a cold side of the stud.

[0318] The coupling assembly of any preceding clause, the nut being coupled to the stud by a second threaded connection.

[0319] A turbine engine comprises a combustion section including a combustion chamber and a coupling assembly comprising, a cold side component, a hot side component, the cold side component and the hot side component together at least partially forming a combustion chamber, a plurality of fastening mechanisms disposed through the cold side component, and a plurality of flexible plates, each of the plurality of flexible plates being circumferentially spaced from each other and extending from a respective one of the plurality of fastening mechanisms to the hot side component and engaging the hot side component to couple the hot side component to the cold side component such that the plurality of fastening mechanisms are insulated from the combustion chamber.

[0320] The turbine engine of any preceding clause, each of the plurality of flexible plates including one or more flexible plate cooling holes that operably direct cooling air through the plurality of flexible plates.

[0321] The turbine engine of any preceding clause, the hot side component comprising a plurality of flanges, each of the plurality of flexible plates engaging a respective one of the plurality of flanges to couple the hot side component to the cold side component.

[0322] The turbine engine of any preceding clause, the plurality of fastening mechanisms comprising a first fastening mechanism and a second fastening mechanism, and the plurality of flexible plates comprise a first flexible plate extending from the first fastening mechanism to the

hot side component and a second flexible plate extending from the second fastening mechanism to the hot side component.

[0323] The turbine engine of any preceding clause, the first fastening mechanism being located upstream of the second fastening mechanism.

[0324] The turbine engine of any preceding clause, the first flexible plate extending from the first fastening mechanism downstream and to the hot side component.

[0325] The turbine engine of any preceding clause, the second flexible plate extending from the second fastening mechanism upstream and to the hot side component.

[0326] The turbine engine of any preceding clause, the hot side component comprising a plurality of flanges, the plurality of flexible plates engaging the plurality of flanges to couple the hot side component to the cold side component.

[0327] The turbine engine of any preceding clause, the plurality of flanges comprising a first flange and a second flange positioned downstream of the first flange, the first flexible plate engaging the second flange and the second flexible plate engaging the first flange.

[0328] The turbine engine of any preceding clause, the plurality of flanges each including a radial flange portion extending substantially radially from the hot side component and an axial flange portion extending substantially axially from the radial flange portion.

[0329] The turbine engine of any preceding clause, the plurality of flexible plates engaging the axial flange portion.

[0330] The turbine engine of any preceding clause, the first flange including a first radial flange portion and a first axial flange portion, the first axial flange portion extending axially downstream from the first radial flange portion.

[0331] The turbine engine of any preceding clause, the second flange including a second radial flange portion and a second axial flange portion, the second axial flange portion extending axially upstream from the second radial flange portion.

[0332] The turbine engine of any preceding clause, further comprising a space defined between the first flange and the second flange, the plurality of flexible plates extending into the space.

[0333] The turbine engine of any preceding clause, the hot side component comprising one or more panels, the one or more panels being coupled to the cold side component by the plurality of flexible plates.

[0334] The turbine engine of any preceding clause, the plurality of flexible plates each comprising a fastening mechanism aperture for receiving a respective fastening mechanism of the one or more fastening mechanisms.

[0335] The turbine engine of any preceding clause, the plurality of flexible plates each comprising a fastening coupling portion that is coupled to a respective fastening mechanism of the one or more fastening mechanisms and a flange coupling portion that is coupled to a respective flange of the plurality of flanges.

[0336] The turbine engine of any preceding clause, the flange coupling portion comprising one or more first projections and one or more second projections that are separated at the flange coupling portion, the one or more first projections and the one or more second projections engaging with a respective flange of the plurality of flanges.

[0337] The turbine engine of any preceding clause, the one or more first projections engaging a hot side of the respective flange, and the one or more second projections engaging a cold side of the respective flange.

[0338] The turbine engine of any preceding clause, the respective flange being press fit between the one or more first projections and the one or more second projections.

[0339] The turbine engine of any preceding clause, the plurality of flexible plates each comprising a first bend defined at the fastening mechanism coupling portion such that the plurality of flexible plates extend radially inward from the fastening mechanism coupling portion.

[0340] The turbine engine of any preceding clause, the plurality of flexible plates each comprising

one or more second bends defined at the flange coupling portion, the one or more first projections and the one or more second projections being bent at the one or more second bends.

[0341] The turbine engine of any preceding clause, the one or more first projections being bent radially outwardly at the one or more second bends, and the one or more second projections being bent radially inwardly at the one or more second bends.

[0342] The turbine engine of any preceding clause, the one or more first projections and the one or more second projections being split into different directions at the one or more second bends.

[0343] The turbine engine of any preceding clause, the plurality of flexible plates each comprising one or more third bends at the one or more first projections and the one or more second projections.

[0344] The turbine engine of any preceding clause, the one or more first projections being bent radially inwardly at the one or more third bends, and the one or more second projections being bent radially outwardly at the one or more third bends.

[0345] The turbine engine of any preceding clause, the plurality of flexible plates each comprising one or more fourth bends at the one or more first projections and the one or more second projections.

[0346] The turbine engine of any preceding clause, the one or more first projections being bent radially outwardly at the one or more fourth bends, and the one or more second projections being bent radially inwardly at the one or more fourth bends.

[0347] The turbine engine of any preceding clause, the one or more first projections and the one or more second projections engaging the axial flange portion at the one or more fourth bends.

[0348] The turbine engine of any preceding clause, the plurality of flexible plates each comprising a first fastening mechanism aperture and a second fastening mechanism aperture.

[0349] The turbine engine of any preceding clause, the first fastening mechanism aperture being located at the fastening mechanism coupling portion, and the second fastening mechanism aperture being located at the flange coupling portion.

[0350] The turbine engine of any preceding clause, the flange coupling portion engaging a cold side of the axial flange portion.

[0351] The turbine engine of any preceding clause, the plurality of flexible plates each including a first bend at the fastening mechanism coupling portion and a second bend at the flange coupling portion.

[0352] The turbine engine of any preceding clause, the flange coupling portion holding the hot side component at the axial flange portion.

[0353] The turbine engine of any preceding clause, the flange coupling portion being secured to the axial flange portion by at least one of brazing, welding, or adhesive.

[0354] The turbine engine of any preceding clause, the plurality of flexible plates being flexible in multiple degrees of freedom such that the hot side component moves axially, radially, or circumferentially with respect to the cold side component.

[0355] The turbine engine of any preceding clause, the first flexible plate being positioned circumferentially adjacent the second flexible plate.

[0356] The turbine engine of any preceding clause, the first flexible plate being positioned circumferentially aft with respect to the second flexible plate.

[0357] The turbine engine of any preceding clause, a space being defined between the cold side component and the hot side component, the plurality of plates being disposed in the space.

[0358] The turbine engine of any preceding clause, the one or more fastening mechanisms being disposed in the space between the cold side component and the hot side component.

[0359] The turbine engine of any preceding clause, the one or more fastening mechanisms comprising a stud and a stud head, the stud head being disposed on a hot side of the cold side component.

[0360] The turbine engine of any preceding clause, the stud head being disposed in the space between the cold side component and the hot side component.

[0361] The turbine engine of any preceding clause, the cold side component being a shell of a liner of the turbine engine, and the hot side component being a heat shield.

[0362] The turbine engine any preceding clause, the cold side component being an annular dome assembly, and the hot side component being a deflector assembly.

[0363] The turbine engine of any preceding clause, the cold side component including one or more cold side component cooling holes that operably direct the cooling air from a cold side of the cold side component to the space between the cold side component and the hot side component.

[0364] The turbine engine of any preceding clause, the one or more cold side component cooling holes being impingement cooling holes.

[0365] The turbine engine of any preceding clause, the hot side component including one or more hot side component cooling holes that operably direct the cooling air from the space to the combustion chamber.

[0366] The turbine engine of any preceding clause, the one or more hot side component cooling holes being film cooling holes.

[0367] The turbine engine of any preceding clause, the one or more fastening mechanisms each comprising a stud extending through the cold side component.

[0368] The turbine engine of any preceding clause, the fastening assembly further comprising a nut that is coupled to the stud on a cold side of the stud.

[0369] The turbine engine of any preceding clause, the nut being coupled to the stud by a second threaded connection.

[0370] A method of operably causing cooling air to flow through a turbine engine. The method comprises operably causing cooling air to flow through a cold side component of the turbine engine, a hot side component being coupled to the cold side component by a plurality of fastening mechanisms, the cold side component and the hot side component together at least partially forming a combustion chamber. The method comprises operably directing the cooling air through one or more flexible plate cooling holes of a plurality of flexible plates, the plurality of flexible plates each being circumferentially spaced from each other and extending a respective one of the plurality of fastening mechanisms to the hot side component and engaging the hot side component to couple the hot side component to the cold side component such that the plurality of fastening mechanisms are insulated from the combustion chamber.

[0371] The method of the preceding clause, further comprising operably directing the cooling air through the hot side component.

[0372] The method of any preceding clause, further comprising operably directing the cooling air through one or more cold side component cooling holes in the cold side component and into a space defined between the cold side component and the hot side component.

[0373] The method of any preceding clause, further comprising operably directing the cooling air from the space through the one or more flexible plate cooling holes of the plurality of flexible plates.

[0374] The method of any preceding clause, further comprising operably directing the cooling air between the plurality of flexible plates and the hot side component.

[0375] The method of any preceding clause, further comprising operably directing the cooling air through one or more hot side component cooling holes in the hot side component and into the combustion chamber.

[0376] The method of any preceding clause, further comprising film cooling the cap by operably directing the cooling air from the hollow interior through the one or more second cap cooling holes and into the combustion chamber.

[0377] The method of any preceding clause, further comprising impingement cooling the hot side component by operably directing the cooling air from the cold side of the cold side component through the one or more cold side component cooling holes and into the space between the cold side component and the hot side component.

[0378] The method of any preceding clause, the cooling air being compressed air from a compressor section of the turbine engine.

[0379] The method of any preceding clause, the cold side component being a shell of a liner of the turbine engine, and the hot side component being a heat shield.

[0380] The method of any preceding clause, the cold side component being an annular dome assembly, and the hot side component being a deflector assembly.

[0381] The method of any preceding clause, the plurality of flexible plates comprising a first flexible plate and a second flexible plate, the method comprising operably directing cooling air through the first flexible plate and the second flexible plate.

[0382] The method of any preceding clause, the plurality of fastening mechanisms comprising a first fastening mechanism and a second fastening mechanism, the plurality of flexible plates comprising a first flexible plate extending from the first fastening mechanism to the hot side component and a second flexible plate extending from the second fastening mechanism to the hot side component.

[0383] The method of any preceding clause, the first fastening mechanism being located upstream of the second fastening mechanism.

[0384] The method of any preceding clause, the first flexible plate extending from the first fastening mechanism downstream and to the hot side component.

[0385] The method of any preceding clause, the second flexible plate extending from the second fastening mechanism upstream and to the hot side component.

[0386] The method of any preceding clause, the hot side component comprising a plurality of flanges, the plurality of flexible plates engaging the plurality of flanges to couple the hot side component to the cold side component.

[0387] The method of any preceding clause, the plurality of flanges comprising a first flange and a second flange positioned downstream of the first flange, the first flexible plate engaging the second flange and the second flexible plate engaging the first flange.

[0388] Although the foregoing description is directed to the preferred embodiments of the present disclosure, other variations and modifications will be apparent to those skilled in the art and may be made without departing from the spirit or the scope of the disclosure. Moreover, features described in connection with one embodiment of the present disclosure may be used in conjunction with other embodiments, even if not explicitly stated above.

Claims

1. A coupling assembly for a turbine engine, the coupling assembly comprising: a cold side component; a hot side component, the cold side component and the hot side component together at least partially forming a combustion chamber; a plurality of fastening mechanisms disposed through the cold side component; and a plurality of flexible plates, each of the plurality of flexible plates being circumferentially spaced from each other and extending from a respective one of the plurality of fastening mechanisms to the hot side component and engaging the hot side component to couple the hot side component to the cold side component such that the plurality of fastening mechanisms is insulated from the combustion chamber.

2. The coupling assembly of claim 1, wherein each of the plurality of flexible plates includes one or more flexible plate cooling holes that operably direct cooling air through the plurality of flexible plates.

3. The coupling assembly of claim 1, wherein the plurality of fastening mechanisms comprises a first fastening mechanism and a second fastening mechanism, and the plurality of flexible plates comprises a first flexible plate extending from the first fastening mechanism to the hot side component and a second flexible plate extending from the second fastening mechanism to the hot side component.

4. The coupling assembly of claim 3, wherein the first fastening mechanism is located upstream of the second fastening mechanism.
5. The coupling assembly of claim 3, wherein the first flexible plate extends from the first fastening mechanism downstream and to the hot side component.
6. The coupling assembly of claim 3, wherein the second flexible plate extends from the second fastening mechanism upstream and to the hot side component.
7. The coupling assembly of claim 1, wherein the hot side component comprises a plurality of flanges, each of the plurality of flexible plates engaging a respective one of the plurality of flanges to couple the hot side component to the cold side component.
8. The coupling assembly of claim 7, wherein each of the plurality of flanges comprises a radial flange portion extending substantially radially from the hot side component and an axial flange portion extending substantially axially from the radial flange portion.
9. The coupling assembly of claim 8, wherein each of the plurality of flexible plates engages a respective axial flange portion.
10. The coupling assembly of claim 7, wherein the plurality of flanges comprises a first flange and a second flange positioned downstream of the first flange, and the plurality of flexible plates comprises a first flexible plate and a second flexible plate, the first flexible plate engaging the second flange and the second flexible plate engaging the first flange.
11. A turbine engine comprising: a combustion section including a combustion chamber; and a coupling assembly comprising: a cold side component; a hot side component, the cold side component and the hot side component together at least partially forming a combustion chamber; a plurality of fastening mechanisms disposed through the cold side component; and a plurality of flexible plates, each of the plurality of flexible plates being circumferentially spaced from each other and extending from a respective one of the plurality of fastening mechanisms to the hot side component and engaging the hot side component to couple the hot side component to the cold side component such that the plurality of fastening mechanisms are insulated from the combustion chamber.
12. The turbine engine of claim 11, wherein each of the plurality of flexible plates includes one or more flexible plate cooling holes that operably direct cooling air through the plurality of flexible plates.
13. The turbine engine of claim 11, wherein the plurality of fastening mechanisms comprises a first fastening mechanism and a second fastening mechanism, and the plurality of flexible plates comprises a first flexible plate extending from the first fastening mechanism to the hot side component and a second flexible plate extending from the second fastening mechanism to the hot side component.
14. The turbine engine of claim 13, wherein the first fastening mechanism is located upstream of the second fastening mechanism.
15. The turbine engine of claim 13, wherein the first flexible plate extends from the first fastening mechanism downstream and to the hot side component.
16. The turbine engine of claim 13, wherein the second flexible plate extends from the second fastening mechanism upstream and to the hot side component.
17. The turbine engine of claim 11, wherein the hot side component comprises a plurality of flanges, each of the plurality of flexible plates engaging a respective one of the plurality of flanges to couple the hot side component to the cold side component.
18. The turbine engine of claim 17, wherein each of the plurality of flanges comprises a radial flange portion extending substantially radially from the hot side component and an axial flange portion extending substantially axially from the radial flange portion.
19. The turbine engine of claim 18, wherein each of the plurality of flexible plates engages a respective axial flange portion.
20. The turbine engine of claim 17, wherein the plurality of flanges comprises a first flange and a

second flange positioned downstream of the first flange, and the plurality of flexible plates comprises a first flexible plate and a second flexible plate, the first flexible plate engaging the second flange and the second flexible plate engaging the first flange.
